Generalized Portfolio Sorts: Prediction and Permanence in Anomaly Alpha[⋆]

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Abstract

Portfolio sorts are widely used to test whether a firm characteristic predicts the cross-section of expected stock returns. We show that the familiar top-minus-bottom portfolio alpha is numerically identical to the coefficient on the sort-assignment variable in a weighted pooled OLS regression on firm-month observations. When permanent firm differences are correlated with sort assignment, this pooled OLS coefficient suffers from omitted-variable bias, so that the sort alpha is a biased estimate of the risk-adjusted return that portfolio sorts are meant to identify. We prove that the bias is alpha-only; factor-loading estimates remain consistent. We further prove that richer benchmark models cannot eliminate this alpha bias. We introduce a Generalized Portfolio Sorts (GPS) framework that recovers the risk-adjusted return. We call this the prediction alpha, and label the difference to the sort alpha the permanence wedge. For book-to-market, an insignificant 27 bp/month sort alpha masks a prediction alpha of 142 bp/month ($t = 4.53$), offset by a -115 bp/month permanence wedge. Across 171 anomalies, depending on the benchmark factor model, 36–42% change significance status under the prediction alpha.

Keywords: Portfolio sorts, Cross-section of expected returns, Prediction alpha and permanence wedge, Panel data methods in asset pricing

JEL classification: G12, C23, C58, C13

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1 Introduction

Portfolio sorts are among the workhorse tools of empirical asset pricing. Researchers sort firms on a characteristic, form a top-minus-bottom portfolio from the extreme groups, and regress the resulting return spread on benchmark factors. The intercept from this time-series regression, the sort alpha, is then interpreted as the abnormal return associated with the sorting characteristic, that is, the expected return spread left unexplained by the benchmark factors.[1] The appeal of the procedure is immediate: the sort alpha appears to summarize, in a single benchmark-adjusted statistic, whether the sorting characteristic separates firms with high abnormal returns from firms with low abnormal returns. But what, exactly, does the sort alpha estimate?

This question is an identification question, not merely a definitional one. Conventional portfolio sorts follow an aggregation-before-estimation design: each month, the procedure collapses thousands of firm-month return observations into a single top-minus-bottom portfolio return, and then estimates alpha and factor loadings from the resulting time series. This grouping is not incidental: portfolios entered asset pricing tests precisely to attenuate the errors-in-variables problem in estimated betas (Blume, 1970; Black et al., 1972; Fama and MacBeth, 1973). We show that the same aggregation that mitigates this measurement problem conceals a distinct identification problem. The standard predictive interpretation of a significant sort alpha, that the sorting characteristic forecasts the cross-section of expected returns after accounting for standard sources of systematic risk,[2] holds only if a strict cross-sectional random-effects (RE) assumption is satisfied. This assumption is never made explicit, because of how the conventional procedure works. In the time-series regression on portfolio returns, the (portfolio-level) residuals are always

[1] In empirical asset pricing, influential studies using the portfolio sorts approach include Jegadeesh and Titman (1993), Sloan (1996), Cooper et al. (2008), Novy-Marx (2013), Fama and French (2015), Hou et al. (2020), and Hsu et al. (2023), among others. The same approach extends well beyond the anomaly literature. Portfolio sorts are widely used in derivative research (Heston et al., 2023), cryptocurrency research (Liu et al., 2022), ESG research (Friewald et al., 2022), household finance (Barber and Odean, 2000; Bali et al., 2023), research on insider trading (Hvide and Meisner Nielsen, 2025), and studies of mutual fund and hedge fund performance (Kacperczyk et al., 2008; Fung et al., 2008).

[2] Cochrane (2011) frames the cross-section of expected returns as a central question in asset pricing. The resulting empirical literature is substantial: Harvey et al. (2016) count hundreds of candidate predictors, Chen and Zimmermann (2022) catalog more than 200 cross-sectional return predictors, and Jensen et al. (2023) assemble a harmonized global database of over 150 such characteristics.

forced to an average of zero. This is a mechanical property of OLS. As a result, researchers have no way to notice, from the portfolio-level regression alone, that an assumption on the individual firm-level is relevant for the identification of the sort alpha, let alone to test for whether this firm-level assumption is met. Because it hides a crucial firm-level condition, the aggregation of firm-level returns into portfolio returns is not a costless solution to the errors-in-variables problem in estimated betas: the same aggregation that mitigates measurement error also removes the firm-level information needed to detect a failure of the identification condition. Our focus in this paper is the object that this concealed condition protects: the abnormal return attributable to the sorting characteristic itself — the quantity the sort alpha is conventionally understood to measure.

The starting point of our analysis is an exact algebraic equivalence between the conventional portfolio sort factor regression and a firm-level panel regression. The conventional procedure follows the aggregation-before-estimation design described above: each month, it collapses the returns of the firms in the extreme portfolios into a single top-minus-bottom portfolio return, then estimates the alpha and factor loadings from the time series of this aggregated return spread. We show that exactly the same alpha and factor loadings can also be obtained without aggregation, by estimating a single weighted pooled OLS regression on the underlying firm-month observations. The key explanatory variable in this firm-level regression is the sort-assignment indicator, which records whether a firm is assigned to the top portfolio rather than the bottom portfolio in each month of the sample period. The pooled OLS regression uses the same firm-level returns, portfolio weights, and benchmark factors as the conventional sort. We formally prove that the coefficient on the sort-assignment indicator exactly matches the conventional sort alpha, and that the equivalence extends to the factor-loading estimates and to the standard errors.

Because the two procedures are the same estimator written in two representations, they share the same econometric properties. Writing the sort alpha as a pooled-panel coefficient makes visible a firm-level exogeneity condition that the conventional factor regression hides: the coefficient on the sort-assignment indicator is free of omitted-variable bias only if the sort-assignment indicator is uncorrelated with the firm-month regression error. Permanent firm-level return components that are not captured by the benchmark factors form part of this regression error. Because pooled OLS imposes a single intercept across all firms, any correlation between these permanent firm-level return components and the sort assignment is not absorbed by the model; it survives in the error term. In economic terms, the RE assumption requires that the firms a characteristic sends to the top portfolio do not differ permanently, in ways the benchmark factors fail to capture, from the firms it sends to the bottom portfolio. When they do, the sort-assignment coefficient absorbs the systematic top-minus-bottom difference in permanent return components leading to a classical omitted-variable bias. The conventional sort alpha then conflates genuine time-series return predictability with a cross-sectional selection effect, and lacks structural identification as a measure of the former.

Because the pre-estimation aggregation to portfolio-level returns removes firm identities from the subsequent time-series regression, the conventional procedure cannot detect, let alone correct, the (hidden) firm-level exogeneity condition it relies on. But our firm-level panel regression, when augmented with firm fixed effects, can: By allowing for a separate intercept for each firm rather than a single intercept shared by all firms, our estimation approach absorbs the permanent firm-level return components that would otherwise contaminate the sort-assignment coefficient. We refer to this firm-level panel regression approach with firm fixed effects as the "Generalized Portfolio Sorts" (GPS) method. Consequently, we term the coefficient on the sort-assignment indicator in this GPS specification the "prediction alpha" (α^{pred}), to distinguish it from the conventional "sort alpha" (α^{sort}) delivered by pooled OLS estimation (and traditional portfolio sorts). The prediction alpha represents the abnormal return associated with the sorting characteristic after removing permanent return differences across firms, and admits the structural predictive interpretation that the sort alpha is conventionally given but does not, on its own, support. We call the difference between the two estimators the "permanence wedge" (B):

$$B = \alpha^{\text{sort}} - \alpha^{\text{pred}}.$$

The permanence wedge is the omitted-variable bias made explicit: it is exact by construction, and it reflects persistent differences in permanent firm-level return components between the firms assigned to the top and bottom portfolios. The wedge has an expected value of zero exactly when the RE assumption holds. The GPS framework does not take a stand on the economic source of the wedge when it differs from zero. It may reflect unmeasured systematic risk, permanent industry composition, persistent mispricing, or some other latent firm attribute correlated with the sorting characteristic. This agnosticism is not a limitation of the framework but a feature of its logic: the sort-on-X design cannot identify the economic source of a composition effect it was never built to detect, but it can still be used to isolate the predictive parameter, α^{pred}, cleanly from that effect. A natural question is whether this identification failure can be resolved simply by adjusting for more risk factors. It cannot. We show, via a set of sufficient orthogonality conditions, that the permanence wedge is largely invariant to the choice of benchmark factor model, and we find empirical support for this near-invariance throughout the paper. The reason is structural rather than incidental: the omitted object behind the permanence wedge is not a missing time-series factor. It is a permanent, firm-level return component in the latent pooled regression, indexed by firms rather than by time, and therefore orthogonal by construction to the time-series variation that market-level factor models are built to absorb. Adding benchmark factors changes the risk adjustment and can shift both the conventional sort alpha and the prediction alpha, but it does not, by itself, impose orthogonality between the sort-assignment indicator and firms' permanent return components. This delivers a pointed conclusion: standard factor-model robustness checks, which compare an anomaly's alpha across various factor models, are diagnostics for whether an alpha survives time-series risk adjustment, not for whether it reflects within-firm predictability. A sort alpha that is stable across benchmark models is not, by itself, evidence of structural identification. Because the permanence wedge is itself invariant to the choice of factor model, an anomaly's apparent robustness across benchmarks can reflect the stability of this bias rather than a genuine risk-adjusted predictive effect. The asymmetry is itself diagnostic: a component that no benchmark factor can move, but that persistent firm-level controls can shrink (Section 4.5), bears the fingerprint of omitted-variable bias, not of an unmeasured priced factor.

The structural orthogonality between the permanent firm-level component and time-series factor variation explains why the identification problem escaped detection despite three decades of scrutiny. A skeptical reader might object that the sort alpha is the benchmark-adjusted payoff of an implementable trading strategy and is therefore a well-defined, real quantity. This argument is tempting, since the sort alpha indeed can be estimated and often is stable across benchmark factor models commonly used in the literature. The argument however completely neglects that standard robustness checks only test for omitted variables within the specific class of market-level factors, that is, time-indexed factor returns. Any persistent firm-level component (which varies cross-sectionally but not over time) is orthogonal to this class of candidate omitted variables. As a result, standard factor-model robustness checks have tested the sort alpha's sensitivity to alternative market-level risk adjustments. We show that the standard analysis is too narrow for answering the question of whether the sorting characteristic identifies cross-sectional return predictability.

In addition, the GPS framework resolves another shortcoming of the conventional portfolio-sorts approach: portfolio sorts are generally limited to a small number of firm characteristics. Cochrane (2011, p. 1061), for instance, notes that while researchers often "sort assets into portfolios based on a characteristic [...] we cannot do this with 27 variables." A related concern is that portfolio sorts are typically restricted to comparisons between the top and bottom groups, discarding the firms in between. This involves a loss of potentially valuable information on the return predictability of a given characteristic. Patton and Timmermann (2010) formalize this truncation concern and propose monotonicity tests that draw on all portfolios. The identification problem we document is orthogonal to that critique: it would persist even if the test used every firm in the cross-section, as the Fama–MacBeth analogue in Appendix C makes explicit, and it survives any choice of sorting granularity (Appendix B). GPS avoids both limitations: because the decomposition is estimated directly on firm-month observations rather than on pre-aggregated portfolio returns, it extends naturally to multiple characteristics estimated jointly (Section 4.5) and can draw on information from the full cross-section of firms rather than only the extreme portfolios.

Related to this, the GPS framework also points toward a constructive response to the identification problem it identifies. Rather than relying solely on firm fixed effects, researchers can restore the random-effects assumption by expanding the characteristic vector to include persistent, observable firm traits, such as industry affiliation or long-run size, that plausibly span the unobserved heterogeneity driving the wedge. When these controls succeed, the permanence wedge shrinks toward zero and pooled estimation again recovers a valid predictive parameter, without sacrificing the cross-sectional power that a fixed-effects specification forgoes. We illustrate this extension in Section 4.5, where GPS is estimated jointly on multiple characteristics; because the decomposition is additive characteristic by characteristic, the same firm-level regression that flags a compositional problem for one characteristic can, by conditioning on others, also help resolve it.

We apply the GPS framework to 171 anomalies from the Chen and Zimmermann (2022) database, using harmonized value-weighted decile sorts. The book-to-market characteristic illustrates both the magnitude of the identification problem and its consequences for inference. The conventional market-model sort alpha for book-to-market is an insignificant 26.7 basis points per month ($t = 1.30$). GPS reveals that this near-zero alpha combines a prediction alpha of 142.0 basis points per month ($t = 4.53$), which is one of the largest within-firm effects in our sample. This effect is nearly fully offset by a permanence wedge of -115.3 basis points per month. The arithmetic exposes which hypothesis the conventional procedure actually tests: the t-test on the sort alpha tests the composite null $\alpha^{\text{pred}} + B = 0$, which the offsetting wedge nearly satisfies, not the predictive null $\alpha^{\text{pred}} = 0$, which the data decisively reject. This finding directly addresses a common objection to fixed-effects estimation: that removing firm-level heterogeneity also removes useful cross-sectional variation, leaving too little power to detect real effects. Here it does the opposite. Isolating the within-firm component does not weaken the evidence for a value effect; it unmasks a strong predictive signal that pooled estimation obscures behind an offsetting composition effect. The conventional sort alpha does not merely understate the strength of book-to-market's predictive content; in this case, it reverses the statistical conclusion entirely.

The same issue is widespread across the anomaly zoo. Replacing the conventional sort alpha with the GPS prediction alpha changes statistical significance status for 36–42% of the 171 anomalies, depending on the benchmark factor model. The changes occur in both directions. For some characteristics, a significant sort alpha loses significance once permanent firm heterogeneity is removed, which points to the sort characteristic not being the ultimate source of the alpha. For others, an insignificant sort alpha becomes significant because the conventional sort alpha had been suppressed by an opposing permanence wedge. Consistent with the structural argument above, the permanence wedge itself is highly stable across standard factor models: across the market model, the Fama–French three-factor model, the Carhart four-factor model, and the Fama–French five-factor model, cross-model correlations of the permanence wedge exceed 0.99.

Our paper contributes to several streams of literature. Most directly, we advance the methodology of portfolio sorts and long-run event studies, building on foundational work by Fama (1998), Lyon et al. (1999), Kothari and Warner (2008), and Cochrane (2011). Cattaneo et al. (2020) analyze the statistical properties of portfolio sorts and provide guidance on optimal portfolio construction. We complement their work by identifying the panel-data exogeneity condition embedded (yet hidden) in the portfolio sorts methodology and by showing how firm fixed effects restore structural identification when that condition fails.

We also contribute to the literature on the "factor zoo" and the reliability of published return predictors (Harvey et al., 2016; McLean and Pontiff, 2016; Harvey, 2017; Chordia et al., 2020; Hou et al., 2020; Chen, 2021). This literature has emphasized concerns about multiple testing, p-hacking, and publication bias. Researchers have pursued various refinements – including machine learning techniques, stochastic discount factor models, and trading-friction adjustments – to address the proliferation of proposed anomalies, while noting that biases related to micro-cap stocks or frequent rebalancing can inflate paper alphas.[3] Our analysis identifies a distinct and complementary concern, one that

[3]For example, see Bryzgalova (2016), Pukthuanthong et al. (2018), Huang et al. (2023), Harvey and Liu (2021), and Novy-Marx and Velikov (2016). Dello-Preite et al. (2025) build on Merton's (1987) model to derive a stochastic discount factor (SDF) that can price both systematic and unsystematic risk, showing that

survives even absent multiple testing or data mining: when a sorting characteristic correlates with persistent firm attributes, pooled estimation introduces an omitted-variable bias that causes the conventional sort alpha to differ from the prediction alpha, for a single, pre-registered anomaly examined in isolation.

Our work also connects to the long-standing debate about whether characteristics predict returns because they proxy for risk or because of mispricing (Daniel and Titman, 1997; Davis et al., 2000). We argue that this debate is logically downstream of the identification question we raise: before interpreting a significant alpha as compensation for risk or as mispricing, one must first know whether the alpha reflects predictive content or permanent firm-level heterogeneity. Furthermore, our GPS framework establishes a direct link between the portfolio sorts method and the literature analyzing whether factor premia are earned from cross-sectional or temporal variation in firm characteristics. For example, Gerakos and Linnainmaa (2018) show that the value premium is specific to variation in the book-to-market ratio that comes from changes in firm size.

Finally, we contribute to the econometrics of panel data in finance. Petersen (2009) demonstrates the importance of appropriate standard-error adjustments in panel regressions, while Gormley and Matsa (2014) emphasize the value of fixed effects for addressing unobserved heterogeneity in corporate finance research. Thompson (2011) develops methods for inference with cross-sectional and temporal dependence. We show that the same econometric considerations apply to portfolio-based asset pricing tests, and we provide a framework for using standard panel regression methods within the portfolio sorts paradigm.

2 Theoretical Framework

In this section, we show that the conventional characteristic-sorted portfolio alpha is exactly a regression coefficient in a weighted pooled OLS regression estimated on the underlying firm-month observations (Proposition 1). This equivalence has a direct econometric

unsystematic risk accounts for over 70% of their SDF's variation. Examples of machine learning techniques used to tame the factor zoo include Feng et al. (2020), Freyberger et al. (2020), Kozak et al. (2020), and Bryzgalova et al. (2025).

consequence: the standard predictive reading of the sort alpha requires a random-effects (RE) assumption. When permanent firm heterogeneity is not orthogonal to the sorting characteristic, the sort alpha does not isolate the within-firm predictive content. The GPS framework therefore identifies the prediction alpha and isolates the permanence wedge as the omitted-variable bias in the sort alpha, making transparent when benchmark-robust anomaly evidence reflects genuine within-firm predictability and when it reflects persistent cross-firm composition.

We proceed as follows. Section 2.1 introduces the GPS framework. Section 2.2 establishes the exact algebraic equivalence between portfolio sorts and weighted pooled OLS and identifies the RE assumption whose failure gives rise to the identification problem at the heart of our analysis. Section 2.3 splits the conventional sort alpha into the prediction alpha and the permanence wedge. Section 2.4 and Section 2.5 characterize what standard factor-model comparisons do and do not reveal about the within-firm and between-firm components, respectively. Section 2.6 discusses the algebraic interpretation of the permanence wedge, its design-based meaning, and the practical reporting implications of the triplet $(\hat{\alpha}^{\mathcal{A}}_{\text{OLS}}, \hat{\alpha}^{\mathcal{A}}_{\text{FE}}, \hat{B}^{\mathcal{A}})$.

2.1 Model Setup

Consider the static firm-level panel model

$$r_{it} \;=\; (\mathbf{x}_{it} \otimes \mathbf{f}_t)\,\boldsymbol{\theta} \;+\; \mu_i \;+\; \varepsilon_{it}, \tag{1}$$

where r_{it} is firm i's excess return in period t, $\mathbf{x}_{it} = [1,\ \mathbf{d}_{i,t-1}]$ contains a constant and lagged sorting variables, $\mathbf{f}_t = [1,\ f_{1t}, \ldots, f_{Kt}]$ contains a constant and K factor returns, and the Kronecker product generates the full set of main effects and interactions. The term μ_i is a time-invariant firm effect and ε_{it} is an idiosyncratic disturbance.

Regression (1) is specified to align with the canonical portfolio-sort framework. The sorting variables in $\mathbf{d}_{i,t-1}$ summarize the portfolio assignments implied by lagged characteristics; the interactions with $\mathbf{f}_t$ generate the differential factor loadings of the sorted portfolios; and the coefficients on $\mathbf{d}_{i,t-1}$ deliver the corresponding long–short alphas.

To build intuition, consider the canonical univariate case with a single factor f_t and a binary portfolio-assignment indicator $D_{i,t-1} \in \{0,1\}$, where $D_{i,t-1} = 1$ denotes assignment to the top portfolio. Setting $\mathbf{d}_{i,t-1} = D_{i,t-1}$ and $\mathbf{f}_t = [1, f_t]$, equation (1) expands to

$$r_{it} = \alpha + \alpha_D D_{i,t-1} + \beta f_t + \beta_D (D_{i,t-1} \times f_t) + \mu_i + \varepsilon_{it}, \tag{2}$$

where α is a common intercept, α_D is the differential intercept for top-portfolio versus bottom-portfolio firms, β is the common factor loading, and β_D is the differential factor loading. The coefficient α_D captures the average return difference between top- and bottom-portfolio firms after adjusting for differential factor exposure. Whether this coefficient recovers the conventional portfolio-sort alpha depends on whether the regression is estimated with the (weighted) pooled OLS estimator or the fixed effects (FE) estimator. We establish this connection in Proposition 1 below, after formalizing the conventional sort procedure.

For now, we note that the pooled OLS and FE estimators do *not* differ in the orthogonality conditions they impose on the idiosyncratic error term ε_{it}. Both rely on the same exogeneity condition vis-à-vis ε_{it}. The distinction lies elsewhere: in addition to that condition, pooled OLS requires the RE assumption that the firm fixed effect μ_i be uncorrelated with the regressors. In contrast, the FE estimator absorbs μ_i and, hence, does not rely on the RE assumption to hold.

2.2 From Portfolio Sorts to Pooled Panel Regressions

Consider a standard univariate portfolio sort based on a lagged firm characteristic $z_{i,t-1}$. In each period t, firms are assigned to a high portfolio H_t and a low portfolio L_t (for example, the top and bottom deciles). Let w_{it} denote firm i's portfolio weight in period t, with $\sum_{i \in H_t} w_{it} = 1$ and $\sum_{i \in L_t} w_{it} = 1$. Common choices are equal weights or value weights based on beginning-of-period market capitalization.

The conventional portfolio-sort procedure computes the following portfolio returns

$$r_t^H = \sum_{i \in H_t} w_{it} r_{it}, \qquad r_t^L = \sum_{i \in L_t} w_{it} r_{it},$$

and then regresses the spread between the two portfolios, $r_t^H - r_t^L$, on K factor returns as follows:

$$r_t^H - r_t^L \;=\; \alpha_{\mathrm{PS}} \;+\; \sum_{k=1}^{K} c_k\, f_{kt} \;+\; u_t^{\mathrm{HML}}, \qquad t = 1, \ldots, T. \tag{3}$$

The intercept $\hat{\alpha}_{\mathrm{PS}}$ of this time-series regression is the conventional portfolio-sort alpha.

Now define the binary portfolio-assignment indicator

$$D_{i,t-1} \;=\; \mathbf{1}\{i \in H_t\},$$

which equals one if firm i is assigned to the high portfolio using the characteristic observed at the end of period $t-1$, and zero otherwise. Restrict the sample to the union of the high and low portfolios in each period and estimate the firm-level regression

$$r_{it} \;=\; a_0 \;+\; \alpha \cdot D_{i,t-1} \;+\; \sum_{k=1}^{K} b_k\, f_{kt} \;+\; \sum_{k=1}^{K} c_k \left(D_{i,t-1} \times f_{kt}\right) \;+\; \varepsilon_{it}, \tag{4}$$

by weighted least squares, assigning each firm-month observation the same weight w_{it} that is used to form the corresponding portfolio return.[4]

Proposition 1 (Equivalence of GPS Model and Portfolio Sorts)**.** (a) *The weighted pooled OLS estimate $\hat{\alpha}_{\mathrm{OLS}}$ from* (4) *equals $\hat{\alpha}_{\mathrm{PS}}$ from* (3)*. More generally, all coefficient estimates coincide: the interaction coefficients $\hat{c}_k$ in* (4) *equal the factor loadings of the long–short spread in* (3).

(b) *For any lag length H, the Driscoll and Kraay (1998) standard error for $\hat{\alpha}_{\mathrm{OLS}}$ equals the Newey–West standard error for $\hat{\alpha}_{\mathrm{PS}}$.*

Proof sketch. Part (a) follows by aggregation. The weighted average of firm-level fitted values within the high and low portfolios reproduces the fitted portfolio returns, and the weighted pooled OLS normal equations collapse to the time-series normal equations for the long–short portfolio. Part (b) follows because the relevant Driscoll–Kraay score sequence is exactly the sequence of cross-sectional weighted averages of the firm-level scores, which coincides with the Newey–West score sequence for the portfolio-level regression. Appendix A provides the full proof and the general multi-portfolio statement.

[4] In (4), α denotes the population coefficient. The pooled WLS estimate is $\hat{\alpha}_{\mathrm{OLS}}$.

□

Proposition 1 is the paper's central theoretical result. It identifies the exact firm-level estimand behind the conventional portfolio-sort alpha: the sort alpha is a pooled-panel regression coefficient estimated on the underlying firm-month observations. Once the sort alpha is recognized as such, the standard econometric distinction between within-firm and between-firm variation applies directly to its interpretation.

Interpreting the conventional portfolio-sort alpha as evidence that the sorting characteristic predicts expected returns therefore requires the RE assumption that permanent firm heterogeneity be orthogonal to sort assignment. In the panel model, that is exactly the condition under which the pooled OLS and FE regression coefficients coincide in expectation. In the portfolio-sort time-series regression (3), the RE assumption requires $E[u_t^{\mathrm{HML}}] = 0$. That is, the factor-adjusted residual of the long–short spread must not carry a nonzero mean from persistent cross-firm composition.[5]

2.3 The GPS Decomposition

Proposition 1 identifies the conventional portfolio-sort alpha as a pooled-panel estimand. Proposition 2 decomposes this estimand into a within-firm component pertinent to the predictive question and a persistent between-firm component.

Proposition 2 (GPS Decomposition)**.** *For any anomaly, factor specification $\mathcal{A}$, weighting scheme, and sample, the conventional portfolio-sort alpha satisfies the exact decomposition*

$$\boxed{\underbrace{\hat{\alpha}^{\mathcal{A}}_{\mathrm{OLS}}}_{\textit{sort alpha}} = \underbrace{\hat{\alpha}^{\mathcal{A}}_{\mathrm{FE}}}_{\textit{prediction alpha}} + \underbrace{\hat{B}^{\mathcal{A}}}_{\textit{permanence wedge}}} \tag{5}$$

Proof sketch. The sample decomposition in (5) is exact by algebraic rearrangement: $\hat{B}^{\mathcal{A}} = \hat{\alpha}^{\mathcal{A}}_{\mathrm{OLS}} - \hat{\alpha}^{\mathcal{A}}_{\mathrm{FE}}$. Under the standard exogeneity condition with respect to ε_{it}, the within transformation removes μ_i and identifies $\hat{\alpha}^{\mathcal{A}}_{\mathrm{FE}}$ as a consistent estimator of the within-firm coef-

[5]The pooled-versus-within tension is not unique to portfolio sorts. For the canonical Fama–MacBeth cross-sectional regression (Fama and MacBeth, 1973), an analogous result holds: the FM slope is also a weighted pooled panel coefficient, so the same decomposition applies. Appendix C states and proves the formal result.

ficient α, regardless of whether the RE assumption holds. By contrast, the pooled OLS regression in (4), and therefore the conventional sort alpha by Proposition 1, absorbs any systematic difference in μ_i across the high and low portfolios into the pooled sort-assignment coefficient. The permanence wedge $\hat{B}^{\mathcal{A}}$ is therefore the pooled-minus-within coefficient difference and is exact in sample by algebra; its interpretation as the component of the sort alpha that reflects persistent cross-firm heterogeneity in the population depends on the failure of the RE assumption. In that case,

$$\operatorname{plim} \hat{\alpha}^{\mathcal{A}}_{\text{OLS}} = \alpha + B_0, \qquad B_0 \equiv E[\mu_i \mid H] - E[\mu_i \mid L].$$

The population wedge B_0 vanishes if and only if the RE assumption holds.[6] □

Interpretation. The decomposition separates the conventional sort alpha into its within-firm component and the wedge from persistent between-firm composition.

The within-firm component $\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$, which we call the *prediction alpha*, admits a predictive interpretation under the standard exogeneity condition: expected returns differ when a firm's characteristic moves it across the sorting threshold.

The between-firm component $\hat{B}^{\mathcal{A}}$, which we call the *permanence wedge*, is the component pooled estimation absorbs from persistent cross-firm composition. Its population counterpart is $B_0 = E[\mu_i \mid H] - E[\mu_i \mid L]$, the cross-sectional difference in firm-level permanent effects between the high and low portfolios. The decomposition itself is exact by algebra; additional assumptions matter only for how the two components are interpreted.

When the permanence wedge is negative, it attenuates the conventional sort alpha relative to the prediction alpha; we refer to this as *suppression*. When it is positive, it amplifies the sort alpha; we refer to this as *inflation*.

Remark 1 (Alpha-only omitted-variable bias). The omitted permanent firm-level component shifts the assignment coefficient in equation (4) but does not distort the factor loadings. By standard omitted-variable arguments, the bias transmitted by μ_i to each pooled OLS coefficient equals the coefficient from the auxiliary regression of μ_i on the

[6] Relative to the within-firm predictive estimand, the permanence wedge represents an omitted-variable bias in the pooled specification.

corresponding regressor, partialled out with respect to all other regressors. Since μ_i is firm-specific and time-invariant, whereas the factor regressors and their interactions with $D_{i,t-1}$ possess nontrivial orthogonal time-series variation, the sole auxiliary regression yielding a non-zero population coefficient is the regression of μ_i on $D_{i,t-1}$, which equals B_0. Consequently, the factor loadings of equation (4) are identified consistently under pooled OLS regardless of whether the RE assumption holds; the omitted-variable bias is confined to the alpha.

Identification comparison. The decomposition itself is exact and does not depend on any exogeneity condition. What does depend on identifying conditions is the interpretation of the prediction alpha as a within-firm predictive relation.

Both estimators rely on the same standard exogeneity condition with respect to the idiosyncratic error ε_{it}. The single additional restriction needed by pooled OLS is the RE assumption that permanent firm heterogeneity be uncorrelated with portfolio assignment. Pooled OLS, and therefore the conventional sort alpha via Proposition 1, requires the RE assumption to isolate the within-firm predictive effect; the firm fixed effects estimator eliminates this concern entirely by absorbing μ_i through differencing.

The implication is straightforward. When the RE assumption holds, the population permanence wedge is zero and the conventional sort alpha converges to the prediction alpha asymptotically. When the RE assumption fails, the conventional sort alpha conflates two distinct components: the within-firm predictive effect and the wedge from persistent cross-firm composition. The prediction alpha, by contrast, recovers the within-firm predictive effect under the standard exogeneity condition. GPS does not create this identification burden; it makes it visible.

Remark (connection to within–between panel decompositions). The GPS decomposition is the portfolio-sort analogue of the standard within–between decomposition in linear panel models (Mundlak, 1978; Wooldridge, 2010). The novelty here is not the existence of a within/between split in panel econometrics. The novelty is that the field's canonical portfolio-sort procedure admits an exact panel representation in the first place.

Proposition 1 is what makes the decomposition relevant for portfolio sorts.

2.4 Factor-Model Sensitivity and Structural Orthogonality

The behavior of the permanence wedge under factor-model adjustment follows directly from its structure. Its population limit $B_0 = E[\mu_i \mid H] - E[\mu_i \mid L]$ is a difference in cross-sectional means of firm-level permanent effects. The quantity is indexed by firms, not by time: it has no time-series variation to be absorbed by time-indexed regressors. Market-level factor models adjust for time-series covariation between the long-short return and factor realizations, and therefore cannot absorb a quantity that is orthogonal to time-series variation by construction. The permanence wedge is therefore largely unreachable by market-level risk adjustment, and exactly unreachable under Condition F.

The remainder of this subsection formalizes this intuition. Adding a factor to the regression shifts both the conventional sort alpha and the prediction alpha through a standard omitted-variable adjustment, but the relevant factor mean differs across the two estimators. For pooled OLS, it is the unconditional factor mean. For the within-firm estimator, it is the factor mean realized over the calendar periods that carry identifying weight in the within-transformed regression. When these two means coincide, the two shifts are equal and the permanence wedge is left unchanged. The following condition makes this requirement precise and provides a sufficient condition for exact factor-model invariance of the permanence wedge.

Condition F (factor-orthogonality of identifying time weights). Let $\mu_{F_k} = T^{-1} \sum_{t=1}^{T} f_{kt}$, and let $\bar{D}_i^w = \left(\sum_{t=1}^{T} w_{it} D_{i,t-1}\right) / \left(\sum_{t=1}^{T} w_{it}\right)$ denote firm i's weight-adjusted average portfolio assignment. Define the time weights

$$\kappa_t^w \equiv \sum_{i=1}^{N_t} w_{it} \, D_{i,t-1}\left(1 - \bar{D}_i^w\right), \qquad \xi_t \equiv \sum_{i=1}^{N_t} w_{it}\left(D_{i,t-1} - \bar{D}_i^w\right).$$

Factor f_k satisfies *Condition F* if

$$\sum_{t=1}^{T} \kappa_t^w (f_{kt} - \mu_{F_k}) = 0, \tag{6}$$

$$\sum_{t=1}^{T} \xi_t f_{kt} = 0. \tag{7}$$

Restriction (6) requires that the calendar periods carrying within-firm switching leverage are uncorrelated with f_k; the second rules out a pure-factor channel that can arise under time-varying value weights and vanishes identically under equal weighting (where $\xi_t = 0$).

Proposition 3 (Factor-Model Invariance of the Permanence Wedge)**.** *Let $\mathcal{A}'$ and $\mathcal{A} = \mathcal{A}' \cup \{k\}$ be two factor specifications differing only by the inclusion of factor f_k*[7] *and its interaction with $D_{i,t-1}$. If f_k satisfies Condition F, then adding f_k to the factor model shifts the conventional sort alpha and the prediction alpha by the same amount. The permanence wedge is therefore invariant:*

$$B^{\mathcal{A}'} = B^{\mathcal{A}}. \tag{8}$$

Remark. Under equal weighting, where Condition F reduces to (6), the condition is also necessary for exact invariance when $\Delta\beta_k \neq 0$: any violation produces a nonzero shift in (8). Under value weighting, offsetting violations of (6) and (7) could in principle preserve invariance, so the two restrictions are jointly sufficient but not individually necessary.

Proof sketch. Adding factor f_k shifts the conventional sort alpha through a standard omitted-variable adjustment: the differential factor exposure of the high-minus-low portfolio multiplied by the factor premium. The prediction alpha shifts by the same omitted-variable logic applied to the within-transformed regression, but the relevant factor mean is the mean realized over the periods that carry identifying weight, augmented by a pure-factor term proportional to $\sum_t \xi_t f_{kt}$ under time-varying value weights.

[7] Without loss of generality, f_k is expressed in a basis orthogonal to the factors in $\mathcal{A}'$ via the Frisch–Waugh–Lovell theorem.

Condition F sets both departures from μ_{F_k} to zero, so the two shifts are equal:

$$\operatorname{plim} \hat{\alpha}_{\text{OLS}}^{\mathcal{A}'} - \operatorname{plim} \hat{\alpha}_{\text{OLS}}^{\mathcal{A}} \;=\; \operatorname{plim} \hat{\alpha}_{\text{FE}}^{\mathcal{A}'} - \operatorname{plim} \hat{\alpha}_{\text{FE}}^{\mathcal{A}} \;=\; \Delta\beta_k \cdot \mu_{F_k},$$

where $\Delta\beta_k = E[\beta_{ki} \mid H] - E[\beta_{ki} \mid L]$ is the differential loading of the high-minus-low portfolio. Subtracting one shift from the other gives $B^{\mathcal{A}'} = B^{\mathcal{A}}$. Appendix A provides the full derivation. □

The economic content of Proposition 3 is straightforward. Factor-model changes operate through a covariance channel: the sort alpha shifts by the portfolio's differential factor exposure times the factor premium. Condition F ensures that the prediction alpha shifts by exactly the same amount, leaving the permanence wedge unchanged. The condition is economically natural for accounting-based characteristics, where portfolio switching is driven by the reporting calendar rather than by contemporaneous factor realizations. For return-based characteristics, such as momentum, Condition F is a substantive restriction – analogous in its role to the standard exogeneity condition – and should be verified rather than assumed.

This distinction has direct implications for how factor-model robustness exercises should be interpreted. Such exercises assess whether an anomaly "survives" risk adjustment, but under Condition F they can shift the prediction alpha while leaving the permanence wedge unchanged. A sort alpha that is stable across benchmark factor models is therefore not, by itself, evidence of robust within-firm predictability.

The following corollary characterizes how the prediction alpha varies across factor specifications, expressing it as a model-free anchor plus a factor-premium adjustment.

Corollary 1 (Factor-Premium Adjustment of the Prediction Alpha)**.** *Let $\hat{\alpha}_{\text{FE}}^{\varnothing}$ denote the prediction alpha from the zero-factor GPS specification ($K = 0$). Suppose Condition F holds for each factor $k \in \mathcal{A}$. Then*

$$\operatorname{plim} \hat{\alpha}_{\text{FE}}^{\varnothing} \;=\; \operatorname{plim} \hat{\alpha}_{\text{FE}}^{\mathcal{A}} \;+\; \sum_{k \in \mathcal{A}} \Delta\beta_k \cdot \mu_{F_k}. \tag{9}$$

That is, the zero-factor prediction alpha equals the factor-adjusted prediction alpha plus the sum of the corresponding factor-premium adjustments.

Proof sketch. The zero-factor fixed effects regression has a single included regressor $\tilde{D}_{i,t-1}$. Adding all K factors and their interactions simultaneously, the multivariate omitted-variable formula distributes linearly because the short model has only one regressor. Under Condition F, the pure-factor channel vanishes and each interaction channel contributes $\Delta\beta_k \cdot \mu_{F_k}$. No orthogonalization of the factor basis is required. Appendix A provides the full proof. □

Corollary 1 establishes that the factor-adjusted prediction alpha can be expressed as a model-free zero-factor anchor minus the corresponding factor-premium adjustment. The permanence wedge, by contrast, has no analogous factor-premium structure: it is a difference in cross-sectional firm means, generated by persistent cross-firm composition rather than by time-series covariation with factor returns. It follows that standard factor-model robustness checks are informative about the prediction alpha but largely uninformative about the permanence wedge — an asymmetry with direct consequences for how anomaly evidence should be read.

2.5 Scope of Factor-Model Comparisons

A central robustness exercise in the anomaly literature compares sort alphas across competing factor specifications (e.g., market model, Fama–French three-factor, Carhart four-factor, and Fama–French five-factor models) and interprets the resulting stability or instability as evidence for or against return predictability. The following corollary collects the implications of Proposition 3 for such tests.

Corollary 2 (Scope of Factor-Model Comparisons)**.** *If Condition F holds for each factor that distinguishes two factor specifications $\mathcal{A}$ and $\mathcal{A}'$, then*

$$B^{\mathcal{A}} = B^{\mathcal{A}'}.$$

Factor-model disagreement maps entirely into the prediction alpha; the permanence wedge lies outside the scope of factor-model comparisons.

Proof. Apply Proposition 3 to each distinguishing factor. □

The corollary has a clear and consequential implication. An anomaly whose conventional sort alpha is heavily driven by the permanence wedge may look extremely robust across factor models precisely because the dominant component is invariant under Condition F. Factor-model stability is therefore not, in itself, evidence of robust within-firm predictability – a conclusion that inverts the standard interpretation of benchmark robustness exercises.

A related implication concerns the pooled versus fixed effects distinction, which matters for point estimates and not merely for standard errors. Factor adjustments can shift both the conventional sort alpha and the prediction alpha substantially while leaving their difference unchanged. The conventional portfolio-sort procedure obscures this invariance; the GPS decomposition makes it transparent.

2.6 Discussion

The permanence wedge admits interpretation at three distinct levels. Algebraically, it is the part of the conventional sort alpha generated by the cross-portfolio difference in permanent firm-level return components. At the level of research design, it is the component of the sort alpha that the sort-on-X procedure cannot attribute to within-firm variation in X. At the level of economic mechanism, the design does not identify whether the wedge reflects omitted risk, industry composition effects, firm lifecycle differences, persistent mispricing, or some other latent firm attribute correlated with X.

Regardless of its economic origin, the sort on X does not identify the permanence wedge as a function of X. The wedge enters the conventional sort alpha because the sorting procedure pools firms with persistently different expected returns, not because the research design has established that within-firm changes in X drive variation in expected returns.

This distinction matters because a return spread generated by sorting on X is not the same thing as evidence that X itself predicts expected returns. The anomaly literature routinely treats the former as establishing the latter. GPS identifies precisely when that inference is warranted and when it is not.

When the conventional sort alpha is attributable entirely to the permanence wedge, the sorting characteristic carries no within-firm predictive content: a change in the characteristic that moves a firm across the sorting threshold does not change that firm's expected return. The sort may still generate a non-zero return spread by passively classifying firms with persistently different expected returns, but that spread is not evidence that within-firm changes in the characteristic predict returns.

Identifying variation and the scope of the prediction alpha. A recurring concern is that the prediction alpha is identified only from the observations that generate within-firm variation in portfolio assignment and therefore rests on a narrow and potentially unrepresentative subset of the cross section. The relevant constraint, however, is induced by the decile discretization itself. Any estimator that uses the same binary sorting indicator $D_{i,t-1}$ as its regressor inherits the same identifying population for the within-firm predictive question. GPS does not create that constraint; it makes visible a constraint the conventional sort already has. In the canonical top-minus-bottom specification, the prediction alpha estimates the return effect of within-firm characteristic movements large enough to cross the sorting boundary, which is precisely the object the conventional sort is designed to target. The multi-characteristic specification of Section 3.5 illustrates how the GPS logic extends beyond this binary threshold to continuous and jointly conditioned variation.

Relation to Fama–MacBeth regressions. The identification problem documented above is not specific to portfolio sorts. Appendix C establishes that the Fama–MacBeth slope is also a weighted pooled panel coefficient and therefore admits the same decomposition identity as the conventional sort alpha. The difference between within-firm predictive content and persistent cross-firm composition is therefore not peculiar to sorted portfolios; GPS makes it explicit in the canonical sort setting.

Practical implication. For empirical work based on portfolio sorts, the relevant diagnostic is not the conventional sort alpha alone. It is the triplet

$$\left(\hat{\alpha}^{\mathcal{A}}_{\mathrm{OLS}}, \hat{\alpha}^{\mathcal{A}}_{\mathrm{FE}}, \hat{B}^{\mathcal{A}}\right).$$

Reporting the triplet side by side makes transparent whether significance derives from within-firm predictive content, persistent cross-firm composition, or both, and it makes clear what standard factor-model comparisons do and do not establish. GPS therefore does not replace the conventional portfolio-sort design. It clarifies what the conventional sort alpha identifies and what it does not.

3 Empirical Results

This section applies the GPS framework to the Chen and Zimmermann (2022) anomaly database covering 176 firm-level characteristics from July 1963 through December 2024. Section 3.1 describes the data and sample construction. Section 3.2 documents that the GPS decomposition materially alters the conclusions drawn from conventional portfolio sorts. Section 3.3 examines the economic structure of the permanence wedge across factor themes, canonical anomalies, and observable cross-anomaly correlates. Section 3.4 tests factor-model invariance of the permanence wedge. Section 3.5 extends the decomposition to settings with multiple characteristics sorted jointly.

3.1 Data and Sample

We source firm-level characteristics from the Chen and Zimmermann (2022) Open Source Asset Pricing database, which provides 176 anomaly sorting variables covering U.S. equities from July 1963 through December 2024. Monthly stock returns are from CRSP.

GPS estimation. For each anomaly $\mathcal{A}$, we estimate the GPS panel regression in equation (4) on the pooled sample of firms assigned to the top and bottom portfolios, $H_t \cup L_t$, using weighted least squares with value-weighted portfolio weights. The OLS intercept

$\hat{\alpha}^{\mathcal{A}}_{\text{OLS}}$ recovers the conventional portfolio-sort alpha by Proposition 1(a). Adding firm fixed effects to the same specification yields the within-firm estimate $\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$; the between-firm component is the residual $\hat{B}^{\mathcal{A}} = \hat{\alpha}^{\mathcal{A}}_{\text{OLS}} - \hat{\alpha}^{\mathcal{A}}_{\text{FE}}$. Decile sorts are the primary specification throughout; quintile sorts are used in Appendix B to test the granularity predictions of the framework. Value-weighted returns are the primary weighting scheme; equal-weighted results, obtained by re-estimating the GPS regression with equal portfolio weights, are reported as robustness checks. Standard errors are Driscoll–Kraay with a Newey–West kernel at lag $H = 3$ months, which Proposition 1(b) shows to be equal to Newey–West standard errors on the corresponding time-series regression.

Inference for the permanence wedge. Statistical significance of the permanence wedge $\hat{B}^{\mathcal{A}} = \hat{\alpha}^{\mathcal{A}}_{\text{OLS}} - \hat{\alpha}^{\mathcal{A}}_{\text{FE}}$ is assessed via a direct HAC standard error for the difference. Both the OLS and FE estimators are computed from the same firm-month observations with the same weights. Let ψ^{OLS}_t and ψ^{FE}_t denote the period-t influence-function contributions to each intercept estimator, constructed from the Driscoll–Kraay score aggregates. The difference sequence $\psi^{\text{diff}}_t = \psi^{\text{OLS}}_t - \psi^{\text{FE}}_t$ captures the period-level contribution to $\hat{B}^{\mathcal{A}}$. Applying the same Newey–West kernel (lag $H = 3$) to $\{\psi^{\text{diff}}_t\}$ yields $\text{Var}(\hat{B}^{\mathcal{A}})$ with the covariance term $\text{Cov}(\hat{\alpha}_{\text{OLS}}, \hat{\alpha}_{\text{FE}})$ correctly accounted for. Rejection of the null hypothesis H_0: $B^{\mathcal{A}} = 0$ is therefore interpreted as evidence of a non-zero population permanence wedge.

Factor models. We estimate the GPS regression under five alternative factor-model specifications: the zero-factor model ($K = 0$), the market model (CAPM), the Fama and French (1993) three-factor model , the Carhart (1997) four-factor model, and the Fama and French (2015) five-factor model. Factor returns are from the Kenneth French Data Library. The Market model serves as the baseline specification, following Jensen et al. (2023): it is historically coherent throughout the sample period, is the most parsimonious risk-adjusted specification, and, as Section 3.4 later documents, leaves the estimated between-firm component largely unchanged relative to richer specifications. Proposition 3 establishes that the population permanence wedge $B^{\mathcal{A}}$ is exactly invariant to factor-model choice when a testable orthogonality condition holds; Section 3.4 documents that $\hat{B}^{\mathcal{A}}$ is empirically near-invariant, consistent with this condition. Cross-model variation therefore falls primarily on the within-firm estimate $\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$, as formalized by Corollary 1. Multi-model comparisons across the four risk-adjusted specifications trace this factor-premium adjustment (the zero-factor model serves as the model-free anchor of Corollary 1 rather than as a comparative benchmark): systematic compression of $\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$ reveals the extent to which the measured prediction alpha depends on the choice of factor specification. The Fama and French (2015) five-factor model (FF 5F) is included in all multi-model tables (Tables 2, 6, and 9) and in the rolling-window factor model invariance (FMI) analysis of Figure 2.

Sample construction. Of the 176 Chen and Zimmermann (2022) characteristics, five fail to produce a valid GPS decomposition under at least one factor model (typically due to insufficient time-series length or singular matrices from insufficient within-firm variation), leaving a final analysis sample of 171 anomalies.[8]

Table 1: GPS Decomposition Summary

	$\hat{\alpha}_{\text{FE}}$ (bp)			$\hat{B}$ (bp)			$\hat{\alpha}_{\text{OLS}}$ (bp)			N significant	
Model	Mean	Med	SD	Mean	Med	SD	Mean	Med	SD	OLS ($\lvert t_{\text{OLS}}\rvert$)	FE ($\lvert t_{\text{FE}}\rvert$)
Market	49.7	38.6	86.3	−12.1	−0.2	84.8	37.6	35.0	39.0	94	92
FF 3F	43.3	30.8	78.5	−12.2	−0.0	83.2	31.2	26.2	44.4	95	79
Carhart 4F	36.4	24.9	83.2	−11.9	−0.6	81.7	24.5	20.9	32.4	71	79
FF 5F	35.4	25.3	78.1	−11.7	0.2	83.1	23.7	20.7	40.3	72	77

The table reports results from a GPS decomposition, $\hat{\alpha}^{\mathcal{A}}_{\text{OLS}} = \hat{\alpha}^{\mathcal{A}}_{\text{FE}} + \hat{B}^{\mathcal{A}}$, across 171 anomalies from the Chen and Zimmermann (2022) database (Section 3.1). Columns are ordered to reflect the decomposition: the prediction alpha $\hat{\alpha}_{\text{FE}}$ (within-firm) plus the permanence bias $\hat{B}$ (between-firm) equals the conventional sort alpha $\hat{\alpha}_{\text{OLS}}$. The last two columns ("N significant: number of anomalies") report the number of OLS specifications with significant predictors, i.e., $|t_{\text{OLS}}| \geq 1.96$ (OLS column) and the number of fixed effects specifications with significant predictors, i.e, $|t_{\text{FE}}| \geq 1.96$ (FE column). The table reports basis points per month. The analysis is based on decile sorts and applies value-weighting.

Table 1 summarizes the GPS decomposition across the analysis sample. Results in the table show that $\text{SD}(\hat{\alpha}_{\text{OLS}}) < \text{SD}(\hat{\alpha}_{\text{FE}})$ under all four factor models. The ratio of the

[8]The five excluded characteristics – Activism2, AgeIPO, FirmAge, NumEarnIncrease, and RDcap – fail to produce a valid GPS decomposition under at least one of the five factor models, typically due to insufficient time-series length or a singular within-firm covariance matrix resulting from near-zero within-firm variation in the sorting indicator.

standard errors of the alphas, $\mathrm{SD}_{\mathrm{FE}}/\mathrm{SD}_{\mathrm{OLS}}$, is about 1.8–2.6, a compression that reflects the strongly negative empirical covariance between $\hat{\alpha}_{\mathrm{FE}}$ and $\hat{B}$ documented in Section 3.2, and is largest under the Carhart (1997) four-factor model specification. Sections 3.2–3.5 examine each component in detail.

Rolling windows. Rolling-window estimates are used in the temporal stability analysis of Section 3.4 (Figure 2). Windows span 15 years (180 months) and are advanced in annual (12-month) steps; over July 1963–December 2024 this yields 47 calendar windows. Publication dates are taken from the Chen and Zimmermann (2022) database.

3.2 Cross-Sectional Decomposition

The GPS decomposition $\hat{\alpha}^{\mathcal{A}}_{\mathrm{OLS}} = \hat{\alpha}^{\mathcal{A}}_{\mathrm{FE}} + \hat{B}^{\mathcal{A}}$ separates every portfolio-sort alpha into the prediction alpha (the within-firm alpha $\hat{\alpha}^{\mathcal{A}}_{\mathrm{FE}}$, estimated via firm fixed effects) and the permanence wedge (the between-firm component $\hat{B}^{\mathcal{A}}$, capturing persistent cross-sectional heterogeneity). This decomposition is exact by construction; the product representation that gives $\hat{B}^{\mathcal{A}}$ its economic content (Proposition 2) holds exactly in the zero-factor baseline and to a close approximation in the implemented factor-adjusted specifications (Section 3.4). The prediction alpha identifies the within-firm coefficient under the standard exogeneity condition (Section 2.3). We apply the decomposition to the 171 anomalies drawn from the 176 characteristics in the Chen and Zimmermann (2022) database and document two findings. First, suppression – the attenuation of $\hat{\alpha}^{\mathcal{A}}_{\mathrm{OLS}}$ by a negative $\hat{B}^{\mathcal{A}}$ – is at least as prevalent as inflation and is quantitatively large precisely among anomalies with the strongest within-firm predictive content. Second, the permanence wedge materially reclassifies the anomaly zoo: roughly a third of anomalies with significant sort alphas lose significance once the within-firm component is isolated, while 29 anomalies with insignificant sort alphas exhibit significant prediction alphas. Section 3.3 examines the economic structure underlying these patterns.

Aggregate decomposition. Table 2 presents the GPS decomposition for the full analysis sample. Panel A reports distributional statistics for all 171 anomalies under each

Table 2: GPS Decomposition Across the Anomaly Zoo

Panel A: All Anomalies

	$\hat{\alpha}_{\text{OLS}}$ (bp)				$\hat{\alpha}_{\text{FE}}$ (bp)				$\hat{B}$ (bp)	
Model	Mean	Med	p_{25}	p_{75}	Mean	Med	p_{25}	p_{75}	Med	%<0
Market	37.6	35.0	14.5	53.1	49.7	38.6	12.7	70.5	−0.2	50.3
FF 3F	31.2	26.2	3.2	52.8	43.3	30.8	10.0	61.2	−0.0	50.3
Carhart 4F	24.5	20.9	4.9	37.8	36.4	24.9	0.1	55.8	−0.6	50.3
FF 5F	23.7	20.7	−1.9	42.2	35.4	25.3	2.9	54.1	0.2	49.7

Panel B: FE-Significant Anomalies ($|t_{\text{FE}}| \geq 1.96$)

		$\hat{\alpha}_{\text{OLS}}$ (bp)				$\hat{\alpha}_{\text{FE}}$ (bp)				$\hat{B}$ (bp)	
Model	N	Mean	Med	p_{25}	p_{75}	Mean	Med	p_{25}	p_{75}	Med	%<0
Market	92	47.6	42.4	26.2	55.2	81.7	64.2	41.8	87.9	−12.6	70.7
FF 3F	79	44.2	36.9	18.3	72.3	77.2	60.1	44.4	95.6	−7.7	64.6
Carhart 4F	79	31.9	28.7	13.7	45.8	66.3	56.0	31.9	95.6	−18.3	69.6
FF 5F	77	32.5	24.3	−1.6	53.4	64.7	53.1	29.5	93.1	−13.2	64.9

The table reports results from a GPS decomposition, $\hat{\alpha}^{\mathcal{A}}_{\text{OLS}} = \hat{\alpha}^{\mathcal{A}}_{\text{FE}} + \hat{B}^{\mathcal{A}}$, across 171 anomalies from the Chen and Zimmermann (2022) database (Panel A) and the subset of anomalies with a significant within-firm estimate ($|t_{\text{FE}}| \geq 1.96$, Panel B). Columns report distributional statistics for the conventional sort alpha ($\hat{\alpha}^{\mathcal{A}}_{\text{OLS}}$), the prediction alpha ($\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$), and the permanence wedge ($\hat{B}^{\mathcal{A}}$). %<0 reports the fraction of anomalies with a negative between-firm component (suppression). The table reports basis points per month. The analysis is based on decile sorts and applies value-weighting.

of the four factor models. Under the market model, the cross-sectional mean of $\hat{\alpha}^{\mathcal{A}}_{\text{OLS}}$ is 37.6 bp/month, while the mean prediction alpha is substantially larger at 49.7 bp/month. The cross-anomaly median of $\hat{B}^{\mathcal{A}}$ is -0.2 bp/month and the suppression–inflation split is 50.3% vs. 49.7%, but the mean $\hat{B}^{\mathcal{A}}$ is -12.1 bp/month, reflects a tail of strongly suppressed anomalies and understates the economic magnitude of the wedge. The standard deviation of $\hat{B}^{\mathcal{A}}$ is 84.8 bp/month, larger than the sort alpha it modifies, and the strongly negative cross-anomaly correlation between $\hat{B}^{\mathcal{A}}$ and $\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$ ($r = -0.90$, documented below) compresses the sort alpha distribution relative to the prediction alpha distribution: $\text{SD}(\hat{\alpha}_{\text{OLS}}) = 39.0$ bp/month versus $\text{SD}(\hat{\alpha}_{\text{FE}}) = 86.3$ bp/month, a ratio of 2.21× that ranges from 1.77× to 2.57× across factor specifications.

Mean OLS alpha declines monotonically from 37.6 bp/month under the market model to 23.7 bp/month under the (FF 5F), across the rows of Panel A, reflecting progressive absorption of the conventional sort alpha by richer benchmarks. Section 3.4 examines this pattern. The permanence wedge is substantially more stable: the fraction of negative $\hat{B}^{\mathcal{A}}$ varies only from 49.7% to 50.3% across all four specifications, a 0.6 percentage-point spread that is consistent with the near-exact factor-model invariance established in Proposition 3. Section 3.4 confirms this stability at the individual anomaly level, documenting cross-model correlations of $\hat{B}^{\mathcal{A}}$ exceeding 0.99.

Suppression is strongest where within-firm predictability is largest. Panel B of Table 2 restricts the sample to anomalies with statistically significant within-firm estimates ($|\hat{t}_{\text{FE}}| \geq 1.96$), retaining 92 anomalies under the market model. Within this subset, the median prediction alpha ($\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$) is 64.2 bp/month, yet the median sort alpha ($\hat{\alpha}^{\mathcal{A}}_{\text{OLS}}$) is only 42.4 bp/month, a gap of 34% that is driven by a strongly negative permanence wedge: the median $\hat{B}^{\mathcal{A}}$ is -12.6 bp/month, and 70.7% of anomalies exhibit suppression. The attenuation is not incidental. Among anomalies with the strongest within-firm predictive content, the permanence wedge is systematically negative, compressing the sort alpha relative to the object it is conventionally interpreted to measure, as discussed next.

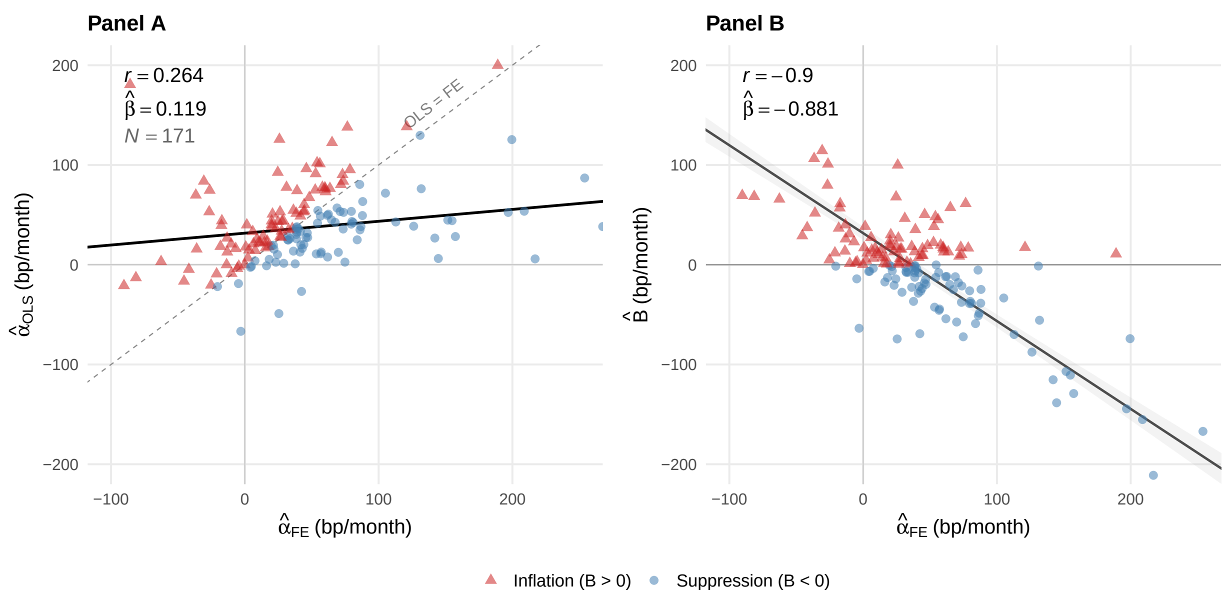

Figure 1: GPS Decomposition: Sort Alpha versus Prediction Alpha. Each point represents one of the 171 anomalies estimated under the market model. Panel A plots the conventional sort alpha ($\hat{\alpha}_{\mathrm{OLS}}$) against the prediction alpha ($\hat{\alpha}_{\mathrm{FE}}$); points below (above) the 45° identity line indicate that the performance wedge is negative (positive), attenuating (inflating) the sort alpha relative to its within-firm component. Panel B plots the permanence wedge ($\hat{B}$) against $\hat{\alpha}_{\mathrm{FE}}$. All estimates use value-weighted decile sorts; quantitative details are in Section 3.2.

Scatter evidence and product structure. Figure 1 displays the decomposition across all 171 anomalies under the market model. In Panel A,the OLS on fixed effects regression slope is 0.119 – far below unity – and the Pearson correlation is only 0.26, indicating that the sort alpha is a poor proxy for the within-firm prediction alpha in the cross-section. The attenuation is a mechanical consequence of the GPS identity: the slope of $\hat{\alpha}^{\mathcal{A}}_{\text{OLS}}$ on $\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$ equals $1 + \text{Cov}(\hat{B}^{\mathcal{A}}, \hat{\alpha}^{\mathcal{A}}_{\text{FE}})/\text{Var}(\hat{\alpha}^{\mathcal{A}}_{\text{FE}})$, which is driven by the strongly negative $\hat{B}$–FE correlation of $r = -0.90$ documented in Panel B.[9] Across the full sample, suppression and inflation are approximately equally prevalent: 86 anomalies (50.3%) exhibit a negative performance wedge and 85 (49.7%) a positive one. The near-even split conceals an important asymmetry, however: among anomalies that are significant under a fixed effects estimator, suppression accounts for 70.7% of cases (Panel B of Table 2)., confirming that the permanence wedge is most consequential precisely where within-firm predictive content is largest.

The product structure introduced in Section 3.3 (exact at $K = 0$; a close approximation in the factor-adjusted specifications) provides the accounting framework for these patterns: $\hat{B}^{\mathcal{A}} \approx (1 - \lambda) \cdot G^{\mathcal{A}}$, where sorting permanence $(1 - \lambda)$ governs *magnitude* and the permanent-return gradient $G^{\mathcal{A}}$ governs *sign*. Under the market model, the median sorting permanence is $(1 - \lambda) = 0.490$: for the typical decile sort, roughly half of portfolio-assignment variation reflects persistent firm-type differences rather than within-firm transitions across the sorting threshold. Sorting permanence is therefore a first-order feature of standard anomaly sorts, not a second-order correction. The median permanent-return gradient is $G^{\mathcal{A}} = -0.3$ bp/month, and thus close to zero, reflecting no systematic tilt in unconditional expected returns across firms with different average portfolio residence. The near-zero median, however, does not imply that $G^{\mathcal{A}}$ is economically unimportant: it is the cross-anomaly *heterogeneity* in $G^{\mathcal{A}}$, not its central tendency, that determines the sign of the permanence wedge for any given anomaly. Where firms persistently assigned to the top portfolio carry lower unconditional expected returns, $G^{\mathcal{A}}$) is negative and the permanence wedge suppresses the sort alpha, where the reverse holds, the wedge inflates it.

[9] Characteristics that generate large prediction alpha tend also to be persistent, inducing sorting permanence that repeatedly assigns the same firms to extreme portfolios and thereby producing a systematic negative covariance between $\hat{B}^{\mathcal{A}}$ and $\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$.

The roughly even 50/50 suppression-inflation split in Panel A follows directly from the near-zero cross-anomaly median of $G^{\mathcal{A}}$). The tilt toward suppression among anomalies that are significant under a fixed effects estimator – 70.7% in Panel B – reflects the empirical regularity that characteristics generating large within-firm predictive effects tend also to produce negative permanent-return gradients, a pattern consistent with the product structure of Proposition 2.

Reclassification of the anomaly zoo. The aggregate statistics in Table 2 establish that the permanence wedge is economically large across the anomaly zoo, but they do not address the question of direct practical relevance: how many anomalies are materially misclassified by the conventional sort alpha? Table 10 in Appendix D reports the GPS decomposition for all 171 anomalies individually. Cross-tabulating sort-alpha significance against prediction-alpha significance at the 5% level yields a four-cell classification that maps the extent of reclassification. Of the 94 anomalies with significant sort alphas, 62 (66%) retain significance under the prediction alpha (Genuine): the sort alpha correctly identifies a within-firm effect, though it typically understates its magnitude. The remaining 32 (34%) lose significance or reverse sign (Compositional): their sort alphas are driven by the permanence wedge. Conversely, 29 anomalies with insignificant sort alphas have significant prediction alphas (Suppressed) – among them book-to-market and several other value-related characteristics whose within-firm predictive content is obscured by an offsetting permance wedge.[10] The Suppressed cell is not populated by marginal cases: it includes some of the most established anomalies in the literature whose within-firm predictive content has been systematically obscured by an offsetting permanence wedge. The remaining 48 anomalies are insignificant under both estimators (*Noise*). Across all four cells, the classification is organized by sorting permanence in a manner consistent

[10] The 5% threshold is applied pointwise to each anomaly. Hence, marginal reclassifications are expected by chance across 171 anomalies. The classification is nonetheless robust to more conservative inference. Tightening the threshold to 1% ($|t| \geq 2.576$) retains 26 of the 32 Compositional anomalies (84%) and 28 of the 29 Suppressed anomalies (97%). The headline cases (book-to-market, operating profitability, and the canonical anomalies examined in Section 3.3) carry t-statistics far from the 1.96 boundary. More fundamentally, the classification is not a threshold artifact: Compositional and Suppressed anomalies have median sorting permanence $1-\lambda$ above 0.68, compared with 0.44 for Genuine anomalies, a structural gradient that reflects the product-structure mechanism of Proposition 2 rather than sampling variation at the significance boundary.

with the product structure of Proposition 2: Compositional and Suppressed anomalies — those for which the permanence wedge materially distorts the sort alpha — have median $1-\lambda$ above 0.68, while Genuine anomalies, for which the sort alpha correctly reflects within-firm predictive content, have median $1-\lambda$ of 0.44. The gradient confirms that sorting permanence is the primary determinant of whether the conventional sort alpha overstates, understates, or correctly measures within-firm predictability.

Inference for the permanence wedge. Using the score-difference HAC procedure described in Section 3.1, the permanence wedge $\hat{B}^{\mathcal{A}}$ is statistically indistinguishable from zero at the 5% level for 77, 80, 79, and 82 of the 171 anomalies under the market model, FF 3F, Carhart, and FF 5F models, respectively (45.0%, 46.8%, 46.2%, and 48.0%). The cross-model overlap is high: of the 87 anomalies significant under at least one specification, 72 (83%) are significant under all four model specifications, and 57 remain significant at the 1% level. The distribution of significance is nearly bimodal: 72 anomalies are robustly significant and 84 are robustly insignificant, with only 15 in between. Thus, evidence against the random-effects restriction is not only broad, but also highly stable across benchmark factor specifications. This stability extends the factor-model-invariance result documented in Section 3.4 from point estimates to inference: changing the benchmark factor model leaves not only the magnitude of the estimated permanence wedge but also its statistical significance largely unchanged, a joint invariance that further supports Condition F as an accurate characterization of the data.

The cross-sectional and reclassification evidence establishes that the GPS decomposition is quantitatively consequential: the permanence wedge is large enough to reverse significance verdicts for more than a third of the anomaly zoo, and evidence against the random-effects restriction is both broad and factor-model stable. Section 3.3 asks whether the wedge has interpretable economic structure, using the product decomposition of Proposition 2 as the organizing framework.

3.3 Economic Structure of the Permanence Wedge

The permanence wedge admits a multiplicative decomposition that organizes its economic interpretation. In the zero-factor case, the wedge factors exactly as

$$\hat{B}^{\mathcal{A}} = \underbrace{(1-\lambda)}_{\text{sorting permanence}} \cdot \underbrace{G^{\mathcal{A}}}_{\text{permanent-return gradient}},$$

where $1-\lambda = \mathrm{Var}(\bar{D}_i)/\mathrm{Var}(D_{i,t-1})$ is the share of total assignment variation attributable to persistent firm-type differences rather than within-firm portfolio transitions, and $G^{\mathcal{A}} = \mathrm{Cov}(\hat{\varphi}_i^{\mathcal{A}}, \bar{D}_i)/\mathrm{Var}(\bar{D}_i)$ is the projection of estimated firm fixed effects onto average portfolio membership – the return premium associated with a unit increase in the fraction of time a firm spends in the top portfolio. Under factor adjustment, the same product representation holds to a close approximation; Appendix A derives the precise conditions under which the approximation is exact.

The aggregate patterns in Table 2 mask substantial variation across economic themes that this product structure predicts: themes whose characteristics are persistent should exhibit larger $|\hat{B}^{\mathcal{A}}|$ than themes whose characteristics are transient. Table 3 tests this prediction at the theme level and Table 4 examines it through four canonical anomalies that span the full range of decomposition patterns.

Theme-level decomposition. We next assign the 170 individual anomalies with a valid theme classification to the 13 factor themes of Jensen et al. (2023). Anomalies are assigned to themes based on their economic category in the Chen and Zimmermann (2022) database, with manual reclassification of anomalies that would otherwise fall into the residual category. As Size is the sole member of a single-anomaly theme, we exclude it, leading to 170 anomalies being assigned to 12 themes. Table 3 reports the median GPS decomposition within each theme. The sharpest contrast is between Value and Momentum. Among the 21 Value anomalies, the median sort alpha of 26.7 bp/month understates the median prediction alpha of 61.9 bp/month by a factor of 2.3 — the highest FE/OLS ratio in the table — and 81% exhibit suppression, with a median permanence wedge, $\hat{B}^{\mathcal{A}}$, of -38.8 bp/month. Both forces in the product structure are at work. Value charac-

Table 3: GPS Decomposition by JKP Factor Theme

Theme	N	Medians (bp/month)				$\hat{B}$<0	FMI	
		$\hat{\alpha}_{\text{OLS}}$	$\hat{\alpha}_{\text{FE}}$	$\hat{B}$	$\hat{\alpha}_{\text{FE}}/\hat{\alpha}_{\text{OLS}}$ (ratio of med.)	% of N	$\rho(\hat{B})$	$\rho(\hat{\alpha}_{\text{FE}})$
Accruals	5	28.7	13.8	−0.8	0.48×	60	0.994	0.955
Debt Issuance	9	44.1	33.8	12.8	0.77×	22	0.997	0.820
Investment	29	26.0	42.9	−12.1	1.65×	69	0.996	0.978
Leverage	5	3.0	18.2	−17.3	6.07×	80	0.998	0.990
Low Risk	36	44.5	55.8	−8.6	1.25×	58	0.995	0.972
Momentum	19	53.1	32.7	10.2	0.62×	21	0.999	0.903
Profit Growth	9	18.3	17.7	6.7	0.97×	33	0.994	0.959
Profitability	16	47.7	29.4	17.6	0.62×	25	0.994	0.905
Quality	6	19.5	1.8	17.7	0.09×	33	0.879	0.871
Seasonality	10	38.5	44.3	3.0	1.15×	50	1.000	0.585
Value	21	26.7	61.9	−38.8	2.32×	81	0.997	0.959
Other	5	3.9	7.6	15.6	1.95×	20	0.998	0.826
All themes	170	35.1	38.7	−0.4	1.10×	51	0.987	0.894

The table reports the GPS decomposition into sort alpha ($\hat{\alpha}^{\mathcal{A}}_{\text{OLS}}$), prediction alpha ($\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$), and permanence wedge ($\hat{B}^{\mathcal{A}}$) aggregated as theme-level medians in basis points per month under the market model, using value-weighted decile sorts. Anomalies are assigned to the 13 factor themes of Jensen et al. (2023) using the economic category variable (`Cat.Economic`) from the Chen and Zimmermann (2022) database, with manual reclassification of anomalies falling into the residual category (e.g., MomSeason* → Seasonality, BetaFP → Low Risk). Size is the sole member of its theme and is excluded, yielding 170 anomalies across 12 reported themes. $\hat{\alpha}_{\text{FE}}/\hat{\alpha}_{\text{OLS}}$ is the ratio of theme-level medians; values above unity indicate that the sort alpha understates the prediction alpha, i.e., the permanence wedge is negative on net. $\hat{B}$<0 (% of N) reports the fraction of anomalies within each theme with a negative permanence wedge. FMI reports mean cross-model Pearson correlations of each GPS component across all six pairwise comparisons of the four factor models, computed for themes with $N \geq 5$; the *All themes* row reports pooled cross-model correlations computed directly on all 170 anomalies rather than as unweighted averages of theme-level FMIs.

teristics, such as book-to-market ratios, are generally highly persistent, generating high sorting permanence $1-\lambda$ and therefore large $|\hat{B}^{\mathcal{A}}|$. The sign of the wedge reflects the permanent-return gradient: firms that persistently reside in the top value decile carry lower unconditional expected returns, producing a negative $G^{\mathcal{A}}$ that attenuates the sort alpha.

The prediction alpha captures a distinct and complementary object — the return effect of within-firm variation in book-to-market, that is, the higher subsequent returns earned by firms that become cheaper relative to their own history. That the conventional sort alpha conflates these two economically distinct sources is precisely what the GPS decomposition is designed to reveal.[11] Momentum presents the opposite picture. Among the 19 Momentum anomalies, the median sort alpha of 53.1 bp/month overstates the median prediction alpha of 32.7 bp/month — an FE/OLS ratio of 0.62× — and the median permanence wedge is positive at +10.2 bp/month. Both features follow directly from the product structure: past-return characteristics are transient by construction, generating low sorting permanence $1-\lambda$ and therefore a small $|\hat{B}^{\mathcal{A}}|$; and the positive permanent-return gradient $G^{\mathcal{A}}$ implies that firms which more frequently populate the top momentum decile carry higher unconditional expected returns, inflating rather than attenuating the sort alpha.

The remaining themes are broadly consistent with the product-structure interpretation. Investment ($N = 29$), whose stock-based characteristics are among the most persistent in the cross-section, exhibits 69% suppression and a median permanence wedge of -12.1 bp/month, reflecting high sorting permanence compressing the sort alpha below the prediction alpha. Themes built on less persistent, flow-based or event-based characteristics (Profitability, Debt Issuance) exhibit smaller $|\hat{B}^{\mathcal{A}}|$ in absolute terms, consistent with lower sorting permanence $(1-\lambda)$ attenuating the magnitude of the permanence

[11] This observation has implications for the use of HML as a risk-adjustment factor. Because HML is constructed from sorted portfolios, its realized return combines the within-firm predictive component of the value premium with the permanent return differential captured by the permanence wedge. When HML is used to risk-adjust other anomalies, the resulting alpha is purged not only of genuine value-factor exposure but also of whatever between-firm composition the permanence wedge reflects. Anomalies with high sorting permanence are particularly susceptible: to the extent that their permanence wedge loads on the same persistent firm characteristics that drive the value wedge, HML adjustment will absorb between-firm return variation that has no structural connection to value-factor risk.

wedge. Their lower suppression rates reflect a tendency toward positive permanent-return gradients $G^{\mathcal{A}}$ rather than low persistence per se: it is the sign of $G^{\mathcal{A}}$, not only its magnitude, that determines whether the wedge suppresses or inflates the sort alpha.[12]

Table 4: GPS Reinterpretation of Canonical Anomalies

Anomaly	Theme	Sort $\hat{\alpha}$ (bp/mo)	Prediction $\hat{\alpha}$ (bp/mo)	Permanence $\hat{B}$ (bp/mo)	$1-\lambda$	Class
BM	Value	26.7	142.0***	−115.3***	0.943	Suppression
ShortInterest	Low Risk	54.4***	54.6***	−0.2	0.789	Neutral
OperProf	Profitability	53.1***	26.3	+26.8*	0.768	Inflation
Mom12m	Momentum	137.9***	76.6***	+61.3***	0.193	Inflation

The table reports the GPS decomposition for four canonical anomalies under the market model (value-weighted decile sorts). Sort $\hat{\alpha}$ is the conventional portfolio-sort alpha $\hat{\alpha}^{\mathcal{A}}_{\text{OLS}}$. Prediction $\hat{\alpha}$ is the within-firm alpha $\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$, estimated by firm fixed effects. Permanence $\hat{B}$ is the between-firm component $\hat{B}^{\mathcal{A}} \equiv \hat{\alpha}^{\mathcal{A}}_{\text{OLS}} - \hat{\alpha}^{\mathcal{A}}_{\text{FE}}$, whose magnitude is governed by sorting permanence and the permanent-return gradient via the product structure of Proposition 2. $1-\lambda$ is sorting permanence, the share of portfolio-assignment variation attributable to persistent firm-type differences. Anomalies are classified as Suppressed ($\hat{B} < 0$), Inflated ($\hat{B} > 0$), or Neutral ($|\hat{B}| < 1$ bp/month). BM = book-to-market; ShortInterest = short interest ratio; OperProf = operating profitability (Novy-Marx, 2013); Mom12m = twelve-month momentum (Jegadeesh and Titman, 1993). Significance for the sort and prediction alphas is based on Driscoll–Kraay standard errors (lag = 3); significance for the permanence wedge is based on the Hausman (1978) specification test of H_0: $\hat{B}^{\mathcal{A}} = 0$. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Canonical anomaly case studies. The theme-level patterns are most informative when grounded in specific anomalies. Table 4 examines four well-known cases – book-to-market, short interest, operating profitability, and twelve-month momentum – selected to span the main decomposition patterns, from strong suppression to strong inflation.

Book-to-market is the starkest illustration of suppression. Its sort alpha of 26.7 bp/month is statistically insignificant ($t = 1.30$), placing BM among the weaker anomalies by conventional standards. The prediction alpha, however, is 142.0 bp/month ($t = 4.53$), one of the largest within-firm effects in the sample. The discrepancy is a mechanical consequence of the product structure. Sorting permanence is $1 - \lambda = 0.943$. Thus.

[12]Quality anomalies ($N = 6$) illustrate the role of the permanent-return gradient independently of suppression. Despite high characteristic persistence – consistent with a large $|\hat{B}^{\mathcal{A}}|$ of 17.7 bp/month relative to a median sort alpha of 19.5 bp/month – the permanence wedge is positive rather than negative: firms persistently assigned to the top quality decile carry higher estimated unconditional expected returns, producing a positive $G^{\mathcal{A}}$ that inflates rather than attenuates the sort alpha. The median prediction alpha of only 1.8 bp/month indicates that virtually all of the sort alpha in this theme is attributable to between-firm composition.

94.3% of portfolio-membership variation reflects persistent firm-type differences rather than within-firm transitions, and firms that persistently reside in the top BM decile carry substantially lower unconditional expected returns (permanent-return gradient $G^{\mathcal{A}} = -122.3$ bp/month). The resulting permanence wedge of -115.3 bp/month nearly exactly offsets the prediction alpha in the sort alpha sum. The conventional sort alpha does not merely understate the value effect, it reverses the statistical conclusion.

The prediction alpha is identified from the approximately 6% of portfolio-membership variation that reflects within-firm transitions across the decile threshold. That the 61-year panel yields a t-statistic of 4.53 from this proportionally small share of variation documents that within-firm movements in book-to-market are both economically large and precisely estimated. More broadly, the prediction alpha for a high-permanence anomaly estimates a specific and well-defined object: the return effect of within-firm characteristic changes large enough to move a firm across the sorting threshold. That estimand may differ from the conventional value premium, which conflates within-firm and between-firm variation by construction. The difference between the two is not a limitation of the prediction alpha; it is precisely what the GPS decomposition is designed to measure.

Operating profitability presents the mirror case. Its sort alpha is 53.1 bp/month ($t = 3.20$), significant at the 1% level, yet the prediction alpha is only 26.3 bp/month ($t = 1.54$), falling short of conventional significance thresholds. The inflation arises from a positive permanent-return gradient: firms that persistently reside in the top profitability decile carry higher unconditional expected returns, generating a permanence wedge of $+26.8$ bp that accounts for more than half of the sort alpha. The within-firm evidence does not support the standard interpretation of the sort alpha as evidence of return predictability; the statistical significance of the conventional estimate is sustained entirely by persistent between-firm composition rather than by within-firm characteristic variation.

Momentum is dominated by within-firm predictability but not immune to inflation. Twelve-month momentum illustrates the moderating role of sorting permanence in the product structure. With $(1 - \lambda) = 0.193$, the lowest in the table, past-return characteristics are transient by construction, and even a large permanent-return gradient ($G^{\mathcal{A}} =$

+318.1 bp/month) is compressed into a permanence wedge of only +61.3 bp/month. The prediction alpha of 76.6 bp/month ($t = 3.19$) is significant, indicating that momentum retains meaningful within-firm predictive content, though this conclusion carries the caveat that the standard exogeneity condition is an identifying assumption rather than a mechanical consequence of lagging for return-based characteristics (Section 2.3). The sort alpha of 137.9 bp/month overstates the within-firm effect by roughly 80%, confirming that even for a low-permanence anomaly the wedge is not negligible. Unlike book-to-market, where the permanence wedge overwhelms the prediction alpha, momentum is dominated by its within-firm component, but the decomposition remains qualitatively material.

Calibration. Not every anomaly is materially affected by the decomposition, and short interest illustrates why.Despite high sorting permanence (0.789), its permanent-return gradient is near zero ($G^{\mathcal{A}} = -0.3$ bp/month), producing a negligible permanence wedge (-0.2 bp). The product structure makes clear why: sorting permanence alone is insufficient to generate a non-zero wedge – both forces must be present. Firms that persistently populate extreme portfolios must also differ systematically in their unconditional expected returns for the wedge to matter. Short interest satisfies the first condition but not the second. The GPS decomposition therefore does not uniformly revise the anomaly zoo; it identifies precisely the anomalies for which the conventional sort alpha is a misleading summary of within-firm predictive content, and confirms where it is not.

Cross-model stability at the anomaly level. The canonical-anomaly reinterpretations in Table 4 are robust across factor specifications. For BM, the permanence wedge ranges from -115.3 bp/month (Market) to -97.5 bp/month (FF 3F and Carhart), a cross-model range of 17.8 bp compared with a level of -115.3 bp. This is the widest range among the four canonical anomalies, and it is precisely what Proposition 3 predicts: when Condition F is not exactly satisfied, residual cross-model shifts are amplified by the magnitude of the differential factor loading $\Delta\beta_k$, which is largest for BM through its HML exposure. Cross-model ranges for the remaining anomalies are substantially tighter (5.2 bp for OperProf, 7.0 bp for Mom12m, 3.7 bp for ShortInterest), which is consistent with their smaller differential factor loadings. Section 3.4 examines this near-invariance systematically across all 171 anomalies.

Observable correlates of the wedge across anomalies. The product structure of Proposition 2 predicts that the permanence wedge is governed by two anomaly-level quantities: sorting permanence $(1-\lambda)$ and the permanent-return gradient $G^{\mathcal{A}}$. To assess how much of the cross-anomaly variation in $\hat{B}^{\mathcal{A}}$ can be attributed to observable between-portfolio composition differences — and which firm attributes are most informative — we regress $\hat{B}^{\mathcal{A}}$ on three families of anomaly-level descriptors. The first family captures industry composition: The L_1 distance) between the FF12 industry-share vectors of D_{10} and D_1, measuring the degree to which the extreme deciles draw from different industries. The second is the differential market loading $\Delta\beta_{\mathrm{MKT}}$ from the time-series spread regression, capturing systematic risk differences between the extreme portfolios. The third family consists of time-averaged D_{10} and D_1 differences in seven persistent firm attributes: log market equity, book-to-market, 12-month prior return, gross profitability (Novy-Marx, 2013), asset growth (Cooper et al., 2008), log panel-entry age, and trailing 36-month return volatility. The first five of which match the SMB, HML, UMD, RMW, and CMA factor themes at the characteristic level. The first five match the SMB, HML, UMD, RMW, and CMA factor themes at the characteristic level. Table 5 reports the results.

The persistent firm-attribute family dominates: it alone accounts for 63% of cross-anomaly variation in $\hat{B}^{\mathcal{A}}$, compared with 3.5% for industry dissimilarity and 0.7% for the market-loading differential; jointly, the three families account for 65%, indicating little incremental explanatory power beyond the firm-attribute family. Within the combined specification, the log-size differential is the strongest individual predictor (49.9 bp per unit with $t = 5.27$), followed by the momentum differential (65.0 bp per unit with $t = 2.31$) and industry dissimilarity (-106 bp per unit with $t = -2.33$). The remaining six differentials are individually indistinguishable from zero. That size and momentum differentials are the most informative observable correlates of $\hat{B}^{\mathcal{A}}$ is consistent with the product structure: anomalies that systematically separate firms by size or past return tend also to separate them by unconditional expected return, generating a non-zero permanent-return gradient even when the sorting characteristic itself has no within-firm predictive content.

A structural interpretation of these coefficients is not warranted. The sort-on-X design under which $\hat{B}^{\mathcal{A}}$ is constructed does not identify the permanence wedge as a function of any of the regressors. The cross-anomaly regression coefficients are descriptive summary statistics that characterize which observable between-portfolio differences covary with the wedge, not estimates of a causal relationship. They do not bear on the paper's central claim, which concerns the decomposition of the conventional sort alpha rather than the economic determinants of the permanence wedge.

Significance verdicts depend critically on the source of the sort alpha. The GPS decomposition does not invalidate any of these anomalies as measures of cross-sectional return differences — a significant sort alpha remains a valid strategy-level intercept. What it clarifies is the source of that significance. For BM, a large within-firm predictive signal is masked by a negative permanence wedge, so the sort alpha understates, and in this case reverses the significance of, the within-firm effect. For a compositional anomaly such as OperProf, the permanence wedge inflates the sort alpha, and the within-firm signal is insignificant on its own. The distinction is material for the question that motivates most of the anomaly literature: whether a characteristic predicts returns within firms, as opposed to merely separating firms with persistently different expected returns. The sort alpha conflates these two sources by construction, answering the predictive question with an error equal to $\hat{B}^{\mathcal{A}}$ (Section 2.3). Table 4 documents that this error is large enough to reverse the statistical conclusion for canonical anomalies on both sides of the decomposition.

3.4 Factor Model Invariance

Proposition 3 establishes that the population permanence wedge $B^{\mathcal{A}}$ is exactly invariant to factor-model choice when Condition F holds: adding or removing a factor shifts both $\hat{\alpha}_{\text{OLS}}$ and $\hat{\alpha}_{\text{FE}}$ by the same amount, $\Delta\beta_k \cdot \mu_{F_k}$ (Proposition 3), leaving $\hat{B}^{\mathcal{A}}$ unchanged. When Condition F is well-supported in the data, this exact population invariance translates

Table 5: Cross-Anomaly Characterization of the Permanence Wedge

	(I)	(II)	(III)	(IV)
Industry dissimilarity	-121.27^{**}			-106.13^{**}
	(-2.02)			(-2.33)
Market-loading differential		-23.06		57.68
		(-1.01)		(1.21)
Log-size differential			51.93^{***}	49.85^{***}
			(5.44)	(5.27)
Book-to-market differential			-6.93	8.81
			(-0.39)	(0.46)
Momentum differential			52.05^{*}	65.00^{**}
			(1.89)	(2.31)
Profitability differential			-0.33	43.92
			(-0.00)	(0.52)
Investment differential			9.07	4.87
			(0.38)	(0.19)
Log-age differential			-15.50	-20.87
			(-0.52)	(-0.69)
Return-volatility differential			729.06^{***}	132.55
			(2.81)	(0.26)
N	171	171	171	171
R^2	0.035	0.007	0.626	0.654
Adj. R^2	0.029	0.001	0.610	0.635

The table reports cross-anomaly OLS regressions of the permanence wedge $\hat{B}^{\mathcal{A}}$ (basis points per month; market benchmark; value-weighted decile sorts) on three families of anomaly-level descriptors summarizing persistent between-portfolio composition differences, with one observation per anomaly ($N = 171$). Each regressor is a time-averaged $D_{10}-D_1$ differential constructed from the decile sort of anomaly $\mathcal{A}$. Industry dissimilarity is the L_1 distance between the FF12 industry-share vectors of D_{10} and D_1, bounded in $[0,1]$. The market-loading differential is $\Delta\beta_{\text{MKT}} = \beta_{D_{10},\text{MKT}} - \beta_{D_1,\text{MKT}}$ from the time-series spread regression. The persistent firm-attribute family consists of value-weighted $D_{10}-D_1$ differences in $\log(\text{ME})$, book-to-market, 12-month prior return, gross profitability (Novy-Marx, 2013), asset growth (Cooper et al., 2008), log years since the firm first appears in the Chen and Zimmermann (2022) panel, and the trailing 36-month return volatility. The size, book-to-market, momentum, profitability, and asset-growth differentials match the SMB, HML, UMD, RMW, and CMA factor themes at the characteristic level. All regressors are expressed in natural decimal units (e.g., a 1% monthly return corresponds to to 0.01). Reported coefficients are in basis points per month per unit of the corresponding regressor. Heteroskedasticity-robust (HC1) t-statistics are reported in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

into empirical near-invariance of the estimated permanence wedge across factor specifications. The cross-sectional implication is sharp: as factor models diverge, cross-model correlations of $\hat{B}^{\mathcal{A}}$ should remain near unity while correlations of $\hat{\alpha}_{\mathrm{OLS}}$ decline, a gradient that holds exactly in population under Condition F and should be clearly visible in the data if the condition is approximately satisfied.

Cross-sectional evidence. Table 6 tests this prediction across all six pairwise comparisons of the four factor models (market model, FF 3F, Carhart, and FF 5F). Panel A reports cross-model Pearson and Spearman correlations of each GPS component across the 171 anomalies. The results confirm the theoretical gradient of Proposition 3. The permanence wedge exhibits near-perfect cross-model agreement: Pearson correlations of $\hat{B}^{\mathcal{A}}$ range from 0.992 (market–Carhart) to 0.999 (FF 3F–FF 5F) across all six model pairs, and the corresponding Spearman correlations are virtually identical (difference $<$0.002 across all pairs), confirming that near-invariance is not driven by leverage points. The within-firm estimate $\hat{\alpha}_{\mathrm{FE}}$ shows moderate cross-model variation (Pearson correlations between 0.941 and 0.974), consistent with the factor-premium adjustments of Corollary 1 shifting the prediction alpha differentially across specifications. Spearman correlations for $\hat{\alpha}_{\mathrm{FE}}$ are systematically lower than the corresponding Pearson values (difference 0.04–0.12), indicating that cross-model alignment of the within-firm alpha is partly inflated by anomalies with very large prediction alphas. The sort alpha $\hat{\alpha}_{\mathrm{OLS}}$ varies most, with Pearson correlations ranging from 0.736 (Carhart–FF 5F) to 0.915 (market–FF 3F). The ordering $\mathrm{cor}(\hat{B}^{\mathcal{A}}) > \mathrm{cor}(\hat{\alpha}_{\mathrm{FE}}) > \mathrm{cor}(\hat{\alpha}_{\mathrm{OLS}})$ holds for all six model pairs without exception under both measures. Sign stability provides further corroboration: the fraction of anomalies with $\mathrm{cor}\hat{B}^{\mathcal{A}} < 0$ ranges from 49.7% to 50.3% across all four factor models, a spread of just 0.6 percentage points.

Panel B translates the correlation evidence into economically interpretable magnitudes. The median absolute deviation (MAD) of $\hat{B}^{\mathcal{A}}$ across model pairs ranges from 0.72 bp/month (FF 3F–Carhart, which differ only in the momentum factor) to 1.90 bp/month (market–FF 5F, which differ in four factors). The corresponding MAD of $\hat{\alpha}_{\mathrm{OLS}}$ ranges from 9.24 to 18.86 bp/month. The median cross-model range, i.e., the

Table 6: Factor Model Invariance: Cross-Model Correlations and Deviations

	Panel A: Cross-Model Correlations						**Panel B: Deviations**		
	Pearson			Spearman			MAD (bp/month)		
Model pair	$\hat{B}$	$\hat{\alpha}_{\text{FE}}$	$\hat{\alpha}_{\text{OLS}}$	$\hat{B}$	$\hat{\alpha}_{\text{FE}}$	$\hat{\alpha}_{\text{OLS}}$	$\hat{B}$	$\hat{\alpha}_{\text{FE}}$	$\hat{\alpha}_{\text{OLS}}$
FF 3F–Carhart	0.998	0.941	0.782	0.998	0.809	0.781	0.72	9.55	9.24
Carhart–FF 5F	0.998	0.942	0.736	0.996	0.789	0.693	1.16	13.99	13.29
FF 3F–FF 5F	0.999	0.973	0.912	0.998	0.887	0.878	0.82	11.12	11.17
Market–FF 3F	0.994	0.974	0.915	0.990	0.924	0.860	1.59	11.12	10.24
Market–Carhart	0.992	0.954	0.802	0.991	0.872	0.766	1.63	14.57	14.47
Market–FF 5F	0.994	0.947	0.778	0.988	0.810	0.644	1.90	20.63	18.86
						Median range	3.02	28.24	27.65
						Range ratio ($\hat{B}/\hat{\alpha}_{\text{OLS}}$)		0.108	

The table reports cross-model Pearson and Spearman correlations of GPS components across 171 anomalies (Panel A) and median absolute deviations of each component across model pairs (Panel B), under value-weighted decile sorts. Model pairs are ordered by factor proximity. Proposition 3 predicts that the population permanence wedge $B^{\mathcal{A}}$ is exactly invariant to factor-model choice when Condition F holds. In Panel B, the median range is the per-anomaly spread between the highest and lowest estimate across all four models. The range ratio is the median across anomalies of each anomaly's $\hat{B}^{\mathcal{A}}$-range to $\hat{\alpha}_{\text{OLS}}$-range ratio, which differs from the ratio of cross-anomaly medians (3.02/27.65 = 0.109) because the median of a ratio does not equal the ratio of medians.

per-anomaly spread from the lowest to the highest estimate across all four models, is 3.02 bp/month for $\hat{B}^{\mathcal{A}}$ versus 27.65 bp/month for $\hat{\alpha}_{\text{OLS}}$, a range ratio of 0.108 (median of per-anomaly ratios, which differs from 3.02/27.65 = 0.109 because the median of a ratio does not equal the ratio of medians). One anomaly, ProbInformedTrading (PIN), is a pronounced outlier with a cross-model permanence-wedge range of 94.6 bp/month — three times larger than the next-widest case — concentrated in the market-to-Carhart transition. This pattern is consistent with factor-correlated decile switching: if firms tend to change PIN deciles during episodes that also produce extreme momentum-factor realizations, Condition F of Proposition 3 would be violated for this anomaly. Excluding PIN leaves the median range virtually unchanged and reduces the cross-sectional standard deviation of the range from 11.6 to 7.1 bp/month (Figure 3). Taken together, the MAD and range evidence deliver a sharp conclusion: moving from a single-factor to a five-factor specification shifts $\hat{\alpha}_{\text{OLS}}$ by a median absolute deviation of 18.9 bp/month but shifts $\hat{B}^{\mathcal{A}}$ by less than 1.9 bp/month. The factor-model debate, insofar as it concerns the overall magnitude of portfolio-sort alphas, is in these benchmark specifications predominantly a debate about how factor models adjust the within-firm component, not about material disagreement over the between-firm component.

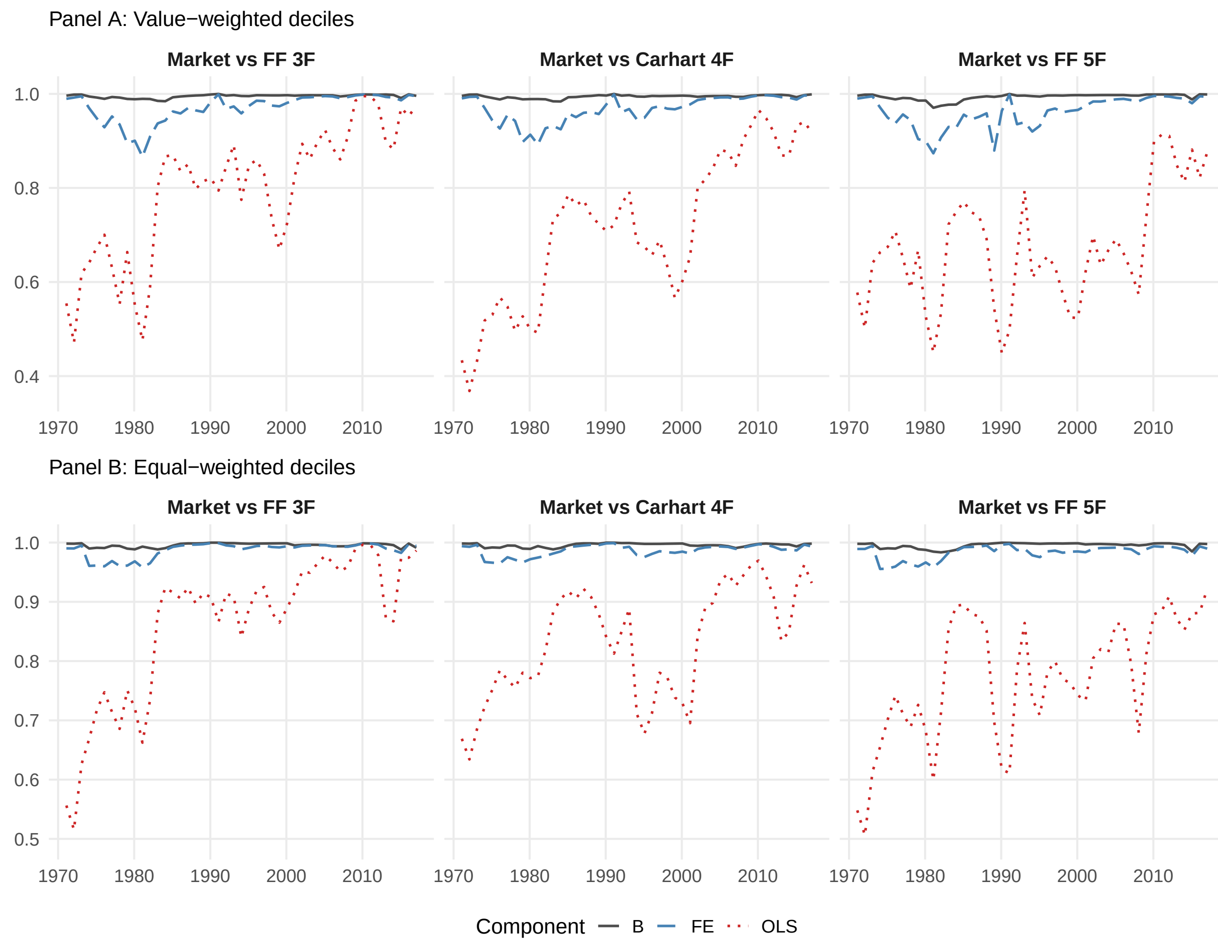

Figure 2: The figure plots cross-model Pearson correlations of $\hat{B}^{\mathcal{A}}$, $\hat{\alpha}_{\text{FE}}$, and $\hat{\alpha}_{\text{OLS}}$ in rolling 15-year windows (180 months, annual step) for three market-anchored model pairs ordered by increasing factor distance: market vs. FF 3F, market vs. Carhart, and market vs. FF 5F. Each rolling window computes the full GPS decomposition independently for every anomaly and correlates the resulting cross-anomaly vectors between model pairs. Panel A reports results for value-weighted decile sorts and Panel B reports results for equal-weighted decile sorts.

Temporal stability. The full-sample evidence in Table 6 establishes that the permanence wedge is near-invariant across factor models, but leaves open whether this stability is confined to a particular sample period or reflects a more fundamental feature of the data. Figure 2 addresses this question by recomputing the GPS decomposition in rolling 15-year windows (180 months, annual step).[13] The theoretical gradient $\text{cor}(\hat{B}^{\mathcal{A}}) > \text{cor}(\hat{\alpha}_{\text{FE}}) > \text{cor}(\hat{\alpha}_{\text{OLS}})$ holds in every value-weighted rolling window for all three market-anchored model pairs — 141 window-pair observations without a single exception — and the same pattern obtains under equal weighting. For the most distant model pair (market vs. FF 5F), the time-averaged rolling correlations are $\text{cor}(\hat{B}^{\mathcal{A}}) = 0.994$, $\text{cor}(\hat{\alpha}_{\text{FE}}) = 0.961$, and $\text{cor}(\hat{\alpha}_{\text{OLS}}) = 0.672$, confirming that the gradient documented in the full sample is stable across the entire time series.

Condition F can also be assessed directly. The switching-weighted factor mean $\omega_{F_k}^{\text{total}} \equiv \sum_t \kappa_t^w f_{kt} / \sum_t \kappa_t^w$ measures the extent to which the calendar periods carrying within-firm switching leverage are representative of the full sample in terms of factor realizations.[14] A ratio $\omega_{F_k}^{\text{total}} / \mu_{F_k}$ close to unity indicates that Condition F is well-approximated in the data. The median ratio across factors and anomalies is 0.999 (IQR: 0.905 to 1.080), strongly consistent with the first restriction in (6) and providing direct empirical support for the near-invariance of $\hat{B}^{\mathcal{A}}$ documented above.

Implication: reframing the factor model debate. The empirical near-invariance of $\hat{B}^{\mathcal{A}}$ across factor models has a direct practical implication. For the purpose of assessing whether an anomaly's significance is materially driven by persistent between-firm composition, the choice of benchmark factor model is largely inconsequential: regardless of which of the four specifications is used, the estimated permanence wedge is nearly the same. This follows from Proposition 2: because $\hat{B}^{\mathcal{A}}$ captures the cross-sectional return differential attributable to persistent between-firm differences rather than within-firm dynamics, it lies largely outside the reach of factor-model adjustment, and researchers need not commit to a particular benchmark to obtain a reliable estimate of it.

When different factor models disagree about whether an anomaly is significant — as they frequently do — the GPS decomposition provides a sharper diagnosis of the source of that disagreement. By Corollary 1, factor-model changes shift the within-firm alpha $\hat{\alpha}_{\text{FE}}$ additively through the factor-premium adjustment, while leaving $\hat{B}^{\mathcal{A}}$ largely unchanged.

[13] Each rolling window computes the full GPS decomposition independently for every anomaly, then correlates the resulting cross-anomaly vectors between model pairs.

[14] κ_t^w is the period-t switching leverage defined in Condition F.

An anomaly that is significant under the market model but insignificant under FF 5F has not been explained away through changes in the between-firm component. The additional factors have shifted the within-firm component, not the permanence wedge. The conventional interpretation, that a sort alpha absorbed by a richer factor model reflects a rewarded risk exposure, pertains to the prediction alpha, not to the permanence wedge, which lies largely outside the reach of factor-model adjustment by construction.

Classification stability across factor models. The near-invariance of the permanence wedge has a sharp implication for how the GPS classification of Section 3.2 behaves across factor models. If the permanence wedge is stable across specifications while the within-firm alpha absorbs factor-premium adjustments, then richer benchmark models should leave the between-firm classification largely unchanged while reclassifying additional anomalies from Genuine to Compositional as their within-firm components are absorbed. Table 7 confirms this prediction: the GPS decomposition yields substantial reclassified shares under every benchmark model, and the fraction of OLS-significant anomalies classified as Compositional rises monotonically as the benchmark factor model becomes richer.

The number of reclassified anomalies — Compositional plus Suppressed — totals 61, 62, 66, and 72 under the market, FF 3F, Carhart, and FF 5F models, respectively. Richer benchmark adjustment does not make the conventional sort procedure more source-diagnostic; if anything, it does the opposite. Among OLS-significant anomalies, the Compositional share rises monotonically from 34% under the market model to 49% under FF 5F: as additional factors absorb the within-firm component of marginal anomalies, the surviving set becomes increasingly dominated by cases whose significance is driven by persistent between-firm composition rather than within-firm predictability. Simultaneously, a sizable and stable set of anomalies are classified as Suppressed across all four specifications — their within-firm components remain significant even after the conventional sort alpha is benchmarked away — confirming that the failure of the conventional sort alpha to reflect within-firm predictive content is not a benchmark-specific artifact.

68 of the 171 anomalies retain the same GPS classification across all four factor models: 24 are always Genuine, 11 always Compositional, 7 always Suppressed, and 26 always Noise. The stable core is economically coherent. Investment and issuance characteristics, including Investment, ShareIss1Y, and ShortInterest, are consistently Genuine, reflecting within-firm predictive content that survives factor-model variation. Profitability characteristics, including GP, OperProfRD, and OrgCap, are consistently Compositional, reflecting sort alphas sustained by persistent between-firm composition across all benchmarks. Several well-established anomalies, including Size and Illiquidity, are consistently Suppressed, their within-firm predictive content masked by an offsetting permanence wedge regardless of the factor specification. That the stable core spans all four classification cells and aligns with recognizable economic themes confirms that the GPS classification reflects the underlying structure of the data rather than a threshold artifact of a particular benchmark.

Table 7: Cross-Model GPS Classification and Permanence-Bias Inference

	Classification counts				Reclassified		$\hat{B}$ inference		
Model	Genuine	Comp.	Suppr.	Noise	N	Share	Sig. 5%	Sig. 1%	Med $\lvert t \rvert$
Market	62	32	29	48	61	35.7%	77	57	1.80
FF 3F	54	41	21	55	62	36.3%	80	61	1.84
Carhart 4F	39	32	34	66	66	38.6%	79	62	1.87
FF 5F	37	35	37	62	72	42.1%	82	63	1.86

The table reports the GPS classification and permanence-wedge inference across four benchmark factor models for all 171 anomalies (value-weighted decile sorts). The left panel reports the four-cell GPS classification based on 5% significance of the conventional sort alpha and the within-firm prediction alpha. An anomaly is classified as *Genuine* if both the sort alpha and the prediction alpha are significant with the same sign, indicating that the conventional sort alpha correctly identifies a within-firm predictive effect. It is classified as *Compositional* if the sort alpha is significant but the prediction alpha is insignificant or opposite-signed, indicating that the sort alpha is driven by persistent between-firm composition rather than within-firm predictability. It is classified as *Suppressed* if the sort alpha is insignificant but the prediction alpha is significant, indicating that a genuine within-firm predictive effect is masked by an offsetting permanence wedge. It is classified as *Noise* if neither alpha is significant. *Reclassified* includes *Compositional* and *Suppressed* anomalies. The right panel reports score-difference HAC inference for the permanence wedge $\hat{B}^{\mathcal{A}} = \hat{\alpha}^{\mathcal{A}}_{\text{OLS}} - \hat{\alpha}^{\mathcal{A}}_{\text{FE}}$, constructed as described in Section 3.1. Sig. 5% and Sig. 1% report the number of anomalies with $|t(\hat{B})| \geq 1.96$ and ≥ 2.576, respectively.

3.5 GPS with Multiple Characteristics

A practical advantage of the GPS framework is that it extends naturally to joint specifications with multiple firm characteristics. For each characteristic j , the additive decomposition $\hat{B}_j = \hat{\alpha}_{j,\text{OLS}} - \hat{\alpha}_{j,\text{FE}}$ is exact, as in the univariate case. The product-structure interpretation of Propositions 2 and 3 extends coefficient by coefficient under the standard exogeneity condition, though this extension proceeds by analogy rather than as an exact replication of the binary-sort equivalence in Proposition 1; the structural decomposition of Corollary 3 continues to hold under the joint normality assumption stated there.[15] The multi-characteristic extension matters practically because conventional higher-order portfolio sorts quickly become sparse: a $5 \times 5 \times 5$ sort on book-to-market, gross profitability, and investment yields a median cell size of only 15 firms, with 27.7% of cells containing fewer than 10; adding a fourth characteristic renders 94.6% of cells sparse. GPS bypasses this dimensionality problem by estimating the corresponding joint specification in a single firm-level regression, without any loss of the decomposition's exactness.

Table 8 reports the three-characteristic joint specification under the market model (1,546,814 firm-month observations; 12,366 firms). The decomposition patterns from the univariate analysis are preserved under joint conditioning, but with informative shifts in magnitude. Book-to-market remains strongly suppressed: its sort alpha of 7.3 bp/month ($t = 0.81$) is statistically insignificant, while its prediction alpha of 43.9 bp/month ($t = 4.89$) is among the largest in the joint specification, with a permanence wedge of -36.5 bp/month. Gross profitability is the most conditioning-sensitive characteristic: its permanence wedge falls by more than half (from $+20.8$ to $+9.6$ bp) once estimated jointly with book-to-market and investment, indicating that part of the univariate between-firm composition effect is attributable to the correlation between profitability and the other two characteristics. Investment displays little sensitivity to joint conditioning, remaining largely prediction-driven in both specifications. The same qualitative patterns persist

[15] The panel model in equation (1) becomes $r_{it} = (\mathbf{d}_{i,t-1} \otimes \mathbf{f}_t)\boldsymbol{\theta} + \mu_i + \varepsilon_{it}$, where $\mathbf{d}_{i,t-1}$ collects the J sorting characteristics, each cross-sectionally standardized to zero mean and unit variance, and $\mathbf{f}_t = [1, f_{1t}, \ldots, f_{Kt}]$ as before. For each characteristic j, the OLS coefficient on $d_{j,i,t-1}$ is the conventional "sort alpha" in a continuous regression sense, and the fixed effects coefficient isolates the within-firm component. In Table 8, the column "OLS $\hat{\alpha}$" refers to the OLS coefficient on each characteristic from the joint regression, not to a literal portfolio-sort intercept.

under the Carhart and Fama–French five-factor models and when momentum is added as a fourth characteristic (Appendix E), confirming that the results are not specific to the market benchmark or the three-characteristic specification. Taken together, these findings establish that the GPS framework scales naturally beyond univariate sorts, delivering an exact decomposition in settings where conventional higher-order portfolio sorts are rendered uninformative by cell sparsity.

Table 8: GPS with Multiple Characteristics

	Univariate continuous GPS			Multi-characteristic GPS		
	OLS $\hat{\alpha}$	Pred $\hat{\alpha}$	$\hat{B}$	OLS $\hat{\alpha}$	Pred $\hat{\alpha}$	$\hat{B}$
Characteristic	(bp)	(bp)	(bp)	(bp)	(bp)	(bp)
Book-to-market	0.8	42.5***	−41.8	7.3	43.9***	−36.5
Gross profitability	24.5*	3.8	20.8	28.9***	19.3	9.6
Investment	13.8	13.9	−0.1	13.1	12.4	0.7

The table reports the GPS decomposition for a three-characteristic joint specification under the market model (1,546,814 firm-month observations; 12,366 firms), comparing the univariate continuous GPS decomposition (left columns) with the corresponding multi-characteristic GPS decomposition (right columns). Both specifications use the same sample and benchmark; the only difference is joint conditioning on the additional characteristics. All characteristics enter as continuous variables, cross-sectionally standardized to zero mean and unit variance each month. OLS $\hat{\alpha}$ is the pooled OLS coefficient on each characteristic, the continuous-regression analogue of the conventional sort alpha. Pred $\hat{\alpha}$ is the corresponding firm fixed-effects coefficient, isolating the within-firm predictive component. $\hat{B} = \hat{\alpha}_{\text{OLS}} - \hat{\alpha}_{\text{FE}}$ is the permanence wedge. A four-characteristic extension adding twelve-month momentum is reported in Appendix E. Significance is based on Driscoll–Kraay standard errors (lag = 3). *, **, ***: 10%, 5%, 1% levels.

4 Conclusion

What does a conventional top-minus-bottom portfolio-sort alpha identify? This paper shows that the answer depends on a firm-level identification condition that is hidden by the conventional aggregation-before-estimation procedure. The standard factor-model regression on monthly portfolio returns is exactly equivalent to a weighted pooled OLS regression on the underlying firm-month observations. In this latent firm-level representation, the conventional sort alpha is the coefficient on top-versus-bottom portfolio assignment. The equivalence extends to the factor-loading estimates and HAC standard errors. Thus, aggregation to portfolio returns changes the representation of the estimator, but not the identification conditions under which its alpha has a predictive interpretation.

The key condition is the random-effects (RE) assumption. Because pooled OLS imposes a common intercept across firms, permanent firm-level return components that are not absorbed by the benchmark factors remain in the regression error. The conventional sort alpha identifies the prediction alpha only if these permanent components are uncorrelated with sort assignment. When the RE assumption fails, the conventional alpha does not isolate the abnormal return associated with the sorting characteristic. It also contains the omitted-variable bias generated by systematic permanent differences between firms assigned to the top and bottom portfolios. The resulting bias affects the alpha, while factor-loading estimates remain consistent.

The Generalized Portfolio Sorts (GPS) framework removes this bias by estimating the same firm-level representation with firm fixed effects. GPS therefore identifies the prediction alpha: the coefficient on sort assignment after permanent firm-level return differences have been absorbed. This object is not external to the portfolio-sort design. It is the alpha that the conventional sort estimates when the random-effects assumption holds. When the RE assumption fails, the difference between the conventional sort alpha and the GPS prediction alpha is the permanence wedge (B),

$$\alpha^{\text{sort}} = \alpha^{\text{pred}} + B.$$

The permanence wedge is the omitted-variable bias in the conventional sort alpha. It measures how much of the conventional alpha is attributable to the correlation between sort assignment and permanent firm-level return components.

Empirically, this distinction is often first-order. For book-to-market, the conventional market-model sort alpha is only 26.7 basis points per month and statistically insignificant. GPS reveals a prediction alpha of 142.0 basis points per month ($t = 4.53$), offset by a permanence wedge of -115.3 basis points per month. Across 171 anomalies from Chen and Zimmermann (2022), replacing the conventional sort alpha with the GPS prediction alpha changes statistical significance status for 36–42% of anomalies, depending on the benchmark model. The changes occur in both directions: some conventional alphas lose significance once permanent firm heterogeneity is removed, while others become significant because the conventional alpha had been suppressed by an opposing permanence wedge. The permanence wedge is also nearly invariant across standard factor models, with cross-model correlations above 0.99, indicating that this identification problem is not generally resolved by adding benchmark factors.

The implication is not that portfolio sorts should be abandoned. It is that the factor-model alpha from a characteristic sort should be interpreted as requiring an explicit firm-level random-effects assumption. Standard factor-model robustness checks remain informative about whether a portfolio-level return spread survives alternative benchmark adjustments. They are much less informative about whether the resulting alpha identifies the predictive component of the sorting characteristic. GPS fills this gap by estimating the prediction alpha directly and by reporting the permanence wedge as the omitted-variable bias embedded in the conventional sort alpha. Before asking whether a significant sort alpha reflects risk compensation or mispricing, one must first know whether the alpha identifies the predictive component that portfolio-sort tests are meant to estimate.

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A Proofs

Proof of Proposition 1 (GPS–Portfolio Sort Equivalence).

We prove the result in full generality: P portfolios, time-varying composition, unbalanced panels, and general weights $w_{it}^{(p)}$ satisfying $\sum_i w_{it}^{(p)} z_{i,t-1}^{(p)} = 1$ for each portfolio p and period t, where $z_{i,t-1}^{(p)} = \mathbf{1}\{i \in p_t\}$. The two-group case stated in Proposition 1 follows by setting $P = 2$ and taking the difference between the H and L portfolios.

Setup. For each portfolio p, the time-series regression is $r_{pt} = \beta_{p,0} + \sum_k \beta_{p,k} f_{kt} + \varepsilon_{pt}$, where $r_{pt} = \sum_i w_{it}^{(p)} z_{i,t-1}^{(p)} r_{it}$. The GPS regression on the pooled firm-level sample with portfolio dummies $\mathbf{d}_{i,t-1} = [z_{i,t-1}^{(1)}, \ldots, z_{i,t-1}^{(P)}]$ and their factor interactions is estimated by WLS with weights $w_{it}^{(p)}$ for firm i belonging to portfolio p in period t. Let $\hat{u}_{it}$ denote the WLS residuals.

Part (a): Coefficients. Define the weighted residual aggregate for portfolio p: $S_t^{(p)} = \sum_i w_{it}^{(p)} z_{i,t-1}^{(p)} \hat{u}_{it}$. Aggregating the firm-level fitted values within portfolio p using the weights $w_{it}^{(p)}$ gives $r_{pt} = \hat{\theta}_{p,0} + \sum_k \hat{\theta}_{p,k} f_{kt} + S_t^{(p)}$. The WLS normal equations for the GPS regression require, for each portfolio p and regressor (intercept and factor f_k):

$$\sum_{i,t} w_{it}^{(p)} z_{i,t-1}^{(p)} \hat{u}_{it} = 0 \quad \text{and} \quad \sum_{i,t} w_{it}^{(p)} z_{i,t-1}^{(p)} f_{kt} \hat{u}_{it} = 0.$$

These reduce to $\sum_t S_t^{(p)} = 0$ and $\sum_t f_{kt} S_t^{(p)} = 0$ for each k, which are the normal equations of the portfolio-level time-series regression with dependent variable r_{pt}. Since both systems share the same dependent variable and orthogonality conditions, they have the same unique solution: $\hat{\theta}_{p,k} = \hat{\beta}_{p,k}$ for all p and k.

For the long–short case ($P = 2$), the HML spread $r_t^H - r_t^L$ aggregates to $\hat{\alpha}_{\text{OLS}} + \sum_k \hat{c}_k f_{kt} + e_t$, where $e_t = S_t^H - S_t^L$. The normal equations $\sum_t e_t = 0$ and $\sum_t f_{kt} e_t = 0$ follow from those for S_t^H and S_t^L, confirming $\hat{\alpha}_{\text{OLS}} = \hat{\alpha}_{\text{PS}}$.

Part (b): Standard errors. The Driscoll and Kraay (1998) covariance estimator for the GPS regression is constructed from the cross-sectional aggregates $h_t = \sum_i w_{it} (\mathbf{d}_{i,t-1} \otimes \mathbf{f}_t)' \hat{u}_{it}$, where $\mathbf{d}_{i,t-1}$ is the portfolio indicator vector and $\mathbf{f}_t$ is the factor row vector from (1). For the portfolio-p block, this aggregate reduces to $S_t^{(p)} \cdot \mathbf{f}_t'$. As shown above, the residual aggregates $S_t^{(p)}$ coincide with the OLS residuals $\hat{\varepsilon}_{pt}$ from the time-series regressions. Hence the Driscoll–Kraay score sequence $\{h_t\}$ is identical to the collection of Newey–

West score sequences $\{\hat{\varepsilon}_{pt} \cdot \mathbf{f}_t'\}$ across all p. With identical bread and meat matrices, the standard errors coincide for each $\hat{\theta}_{p,k}$.

For the long–short case, the score sequence is $\{e_t \cdot \mathbf{f}_t'\}$, which matches the Newey–West score for $\hat{\alpha}_{\text{PS}}$ from (3). $\square$

Proof of Proposition 2 (GPS Decomposition).

For any factor specification $\mathcal{A}$ and weighting scheme, the pooled OLS coefficient on portfolio assignment $\hat{\alpha}^{\mathcal{A}}_{\text{OLS}}$ and the firm-fixed-effects coefficient $\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$ are computed from the same sample. Their difference $\hat{B}^{\mathcal{A}} \equiv \hat{\alpha}^{\mathcal{A}}_{\text{OLS}} - \hat{\alpha}^{\mathcal{A}}_{\text{FE}}$ is an algebraic identity that holds without distributional or exogeneity assumptions. This establishes the exact decomposition in (5). $\square$

Derivation of the Product Structure of the Permanence Bias (used in Section 3.3).

Zero-factor case ($K = 0$). Without factors, the GPS regression reduces to a simple pooled WLS of r_{it} on a constant and $D_{i,t-1}$. For a single-regressor panel regression, pooled OLS is a scalar-weighted average of the within and between estimators (see, e.g., Wooldridge, 2010, Ch. 10):

$$\hat{\alpha}^{\mathcal{A}}_{\text{OLS}} = \lambda\, \hat{\alpha}^{\mathcal{A}}_{\text{FE}} + (1-\lambda)\, \hat{\alpha}^{\mathcal{A}}_{B},$$

where $\lambda = \text{Var}_W(D_{i,t-1})/\text{Var}(D_{i,t-1})$.

In the zero-factor case, the standard within-between variance identity gives $\text{Var}(D) = \text{Var}_W(D) + \text{Var}(\bar{D})$, so $1 - \lambda = \text{Var}(\bar{D}_i)/\text{Var}(D_{i,t-1})$, consistent with the main-text definition. Subtracting $\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$ from both sides:

$$\hat{B}^{\mathcal{A}} = (1-\lambda)\big(\hat{\alpha}^{\mathcal{A}}_{B} - \hat{\alpha}^{\mathcal{A}}_{\text{FE}}\big) = (1-\lambda)\, G^{\mathcal{A}},$$

where $G^{\mathcal{A}} = \text{Cov}(\hat{\varphi}^{\mathcal{A}}_i, \bar{D}_i)/\,\text{Var}(\bar{D}_i)$ is the permanent-return gradient and $\hat{\varphi}^{\mathcal{A}}_i$ is the estimated firm fixed effect. This product structure is exact.

Factor case ($K > 0$). Fix a factor specification $\mathcal{A}$. By the Frisch–Waugh–Lovell theorem, partialling out the included factor terms from both returns and portfolio membership leaves the *pooled OLS* coefficient on the residualized portfolio-membership regressor unchanged. For that residualized single-regressor panel regression, pooled OLS is a scalar-weighted average of the within and between estimators:

$$\hat{\gamma}^{\mathcal{A}}_{\text{OLS}} = \lambda^{\mathcal{A}}\, \hat{\gamma}^{\mathcal{A}}_{\text{within}} + (1-\lambda^{\mathcal{A}})\, \hat{\gamma}^{\mathcal{A}}_{\text{between}},$$

where $\lambda^{\mathcal{A}} = \text{Var}_W(\tilde{D}^{\mathcal{A}}_{i,t-1})/\text{Var}(\tilde{D}^{\mathcal{A}}_{i,t-1})$ is the within-variance share of the FWL-residualized portfolio indicator. By FWL, $\hat{\gamma}^{\mathcal{A}}_{\text{OLS}} = \hat{\alpha}^{\mathcal{A}}_{\text{OLS}}$. However, the within estimator on pooled-OLS-residualized data, $\hat{\gamma}^{\mathcal{A}}_{\text{within}}$, does not in general equal $\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$ from the full multiple regression, because the pooled OLS projection operators used in the FWL residualization do not commute exactly with the within transformation.

Defining $\hat{B}^{\mathcal{A}}_{\text{FWL}} \equiv \hat{\gamma}^{\mathcal{A}}_{\text{OLS}} - \hat{\gamma}^{\mathcal{A}}_{\text{within}}$, the single-regressor within–between decomposition gives

$$\hat{B}^{\mathcal{A}}_{\text{FWL}} = (1 - \lambda^{\mathcal{A}}) \cdot \underbrace{\left(\hat{\gamma}^{\mathcal{A}}_{\text{between}} - \hat{\gamma}^{\mathcal{A}}_{\text{within}}\right)}_{\equiv\ \hat{\Delta}^{\mathcal{A}}},$$

which is exact in the residualized problem. At $K = 0$, $\hat{\gamma}^{\mathcal{A}}_{\text{within}} = \hat{\alpha}^{\mathcal{A}}_{\text{FE}}$ exactly, so $\hat{B}^{\mathcal{A}}_{\text{FWL}} = \hat{B}^{\mathcal{A}}$, $\lambda^{\mathcal{A}}$ coincides with λ, and the between-minus-within gap equals the permanent-return gradient $G^{\mathcal{A}} \equiv \text{Cov}(\hat{\varphi}_i^{\mathcal{A}}, \bar{D}_i) / \text{Var}(\bar{D}_i)$, recovering the exact zero-factor product representation introduced in Section 3.3. At $K > 0$, the non-commutation introduces a discrepancy $\hat{B}^{\mathcal{A}} - \hat{B}^{\mathcal{A}}_{\text{FWL}} = \hat{\gamma}^{\mathcal{A}}_{\text{within}} - \hat{\alpha}^{\mathcal{A}}_{\text{FE}}$, so the zero-factor product representation $\hat{B}^{\mathcal{A}} \approx (1 - \lambda)\, G^{\mathcal{A}}$ is an approximation. Section 3.4 documents near-invariance of $\hat{B}^{\mathcal{A}}$ across factor models, consistent with this discrepancy being small in practice. □

Proof of Proposition 3 (Factor-Model Invariance of the Permanence Bias).

Let $\mathcal{A} = \mathcal{A}' \cup \{k\}$, so that $\mathcal{A}$ includes factor f_k and the interaction $D_{i,t-1} \times f_{kt}$, whereas $\mathcal{A}'$ omits both.

OLS sensitivity. At the portfolio level, the H-L spread under $\mathcal{A}$ includes the term $c_k f_{kt}$ in the time-series regression, where $c_k = \Delta\beta_k$ by Proposition 1. Omitting f_{kt} from the HML time-series regression shifts the intercept by the mean of the omitted term:

$$\text{plim}\ \hat{\alpha}^{\mathcal{A}'}_{\text{OLS}} - \text{plim}\ \hat{\alpha}^{\mathcal{A}}_{\text{OLS}} = c_k\, \mu_{F_k} = \Delta\beta_k\, \mu_{F_k}.$$

Within-firm shift. Omitting factor f_k and its interaction from the FE regression shifts the within-firm alpha through two channels: (i) an *interaction channel*, through the covariance of the within-transformed indicator with $D_{i,t-1}\, f_{kt}$; and (ii) a *pure-factor channel*, through the covariance of the within-transformed indicator with f_{kt} itself. We derive the closed-form expressions under equal weighting, where the pure-factor channel vanishes and the interaction channel yields a transparent closed form, and then extend to value weights.

Equal-weight case. In the within-transformed regression, the demeaned indicator is $\tilde{D}_{i,t-1} = D_{i,t-1} - \bar{D}_i$.

Pure-factor channel vanishes. Since the number of firms in the top portfolio is constant over time ($\sum_i D_{i,t-1} = N_H$ for all t),

$$\sum_{i=1}^{N} \bar{D}_i = T^{-1} \sum_{t=1}^{T} \sum_{i=1}^{N} D_{i,t-1} = N_H,$$

so for every t,

$$\sum_{i=1}^{N} \tilde{D}_{i,t-1} = \sum_{i=1}^{N} D_{i,t-1} - \sum_{i=1}^{N} \bar{D}_i = 0.$$

Since f_{kt} is common across firms within period t,

$$\sum_{i,t} \tilde{D}_{i,t-1} \, f_{kt} = \sum_{t} f_{kt} \sum_{i} \tilde{D}_{i,t-1} = 0.$$

The pure factor term is therefore orthogonal to $\tilde{D}_{i,t-1}$, so channel (ii) contributes no shift. In the notation of Condition F, $\xi_t = \sum_i \tilde{D}_{i,t-1} = 0$ for all t, so (7) holds automatically under equal weighting.

Interaction channel. The entire FE shift comes from omitting the interaction $D_{i,t-1} f_{kt}$. By the omitted-variable formula,

$$\operatorname{plim} \hat{\alpha}_{\mathrm{FE}}^{\mathcal{A}'} - \operatorname{plim} \hat{\alpha}_{\mathrm{FE}}^{\mathcal{A}} = c_k \cdot \frac{\sum_{i,t} \tilde{D}_{i,t-1} \, (\widetilde{D_{i,t-1} f_{kt}})_i}{\sum_{i,t} \tilde{D}_{i,t-1}^2}.$$

Since $\sum_t \tilde{D}_{i,t-1} = 0$ for each i,

$$\sum_{i,t} \tilde{D}_{i,t-1} \, (\widetilde{D_{i,t-1} f_{kt}})_i = \sum_{i,t} \tilde{D}_{i,t-1} \, D_{i,t-1} \, f_{kt}.$$

Since $D_{i,t-1} \in \{0,1\}$, we have $\tilde{D}_{i,t-1} \, D_{i,t-1} = D_{i,t-1}(1 - \bar{D}_i)$. The numerator therefore becomes

$$\sum_{i,t} \tilde{D}_{i,t-1} \, D_{i,t-1} \, f_{kt} = \sum_{t=1}^{T} f_{kt} \sum_{i=1}^{N} D_{i,t-1}(1 - \bar{D}_i) = \sum_{t=1}^{T} \kappa_t \, f_{kt},$$

where $\kappa_t \equiv \sum_{i=1}^{N} D_{i,t-1}(1 - \bar{D}_i)$ is the switching leverage in period t. For the denominator, since $D_{i,t-1} \in \{0,1\}$, summing the squared within-transformed indicator over time yields

$$\sum_{t=1}^{T} \tilde{D}_{i,t-1}^2 = \sum_{t=1}^{T} \left(D_{i,t-1} - \bar{D}_i \right)^2 = T \, \bar{D}_i (1 - \bar{D}_i) = \sum_{t=1}^{T} D_{i,t-1}(1 - \bar{D}_i),$$

so

$$\sum_{i=1}^{N} \sum_{t=1}^{T} \tilde{D}_{i,t-1}^{2} = \sum_{t=1}^{T} \kappa_t.$$

Combining numerator and denominator,

$$\text{plim } \hat{\alpha}_{\text{FE}}^{\mathcal{A}'} - \text{plim } \hat{\alpha}_{\text{FE}}^{\mathcal{A}} = \Delta\beta_k \cdot \frac{\sum_{t=1}^{T} \kappa_t f_{kt}}{\sum_{t=1}^{T} \kappa_t} = \Delta\beta_k \cdot \omega_{F_k},$$

where $\omega_{F_k} \equiv \sum_t \kappa_t f_{kt} \big/ \sum_t \kappa_t$ is the switching-weighted factor mean. Writing $W_\kappa \equiv \sum_t \kappa_t$ for the total switching leverage and using $\omega_{F_k} = \mu_{F_k} + S_\kappa / W_\kappa$, the equal-weight FE shift decomposes as

$$\text{plim } \hat{\alpha}_{\text{FE}}^{\mathcal{A}'} - \text{plim } \hat{\alpha}_{\text{FE}}^{\mathcal{A}} = \underbrace{\Delta\beta_k \cdot \mu_{F_k}}_{\text{OLS shift}} + \underbrace{\Delta\beta_k \cdot S_\kappa / W_\kappa}_{\text{interaction departure}}, \tag{10}$$

where $S_\kappa \equiv \sum_t \kappa_t (f_{kt} - \mu_{F_k})$. Condition (7) holds automatically under equal weighting (since $\xi_t = 0$ for all t). Condition (6) sets $S_\kappa = 0$, eliminating the interaction departure and making the FE shift equal the OLS shift.

Extension to value weights. Under value weighting, the within transformation uses the weight-adjusted mean $\bar{D}_i^w \equiv \left(\sum_{s=1}^{T} w_{is} D_{i,s-1}\right) \big/ \left(\sum_{s=1}^{T} w_{is}\right)$ and the demeaned indicator $\tilde{D}_{i,t-1}^w \equiv D_{i,t-1} - \bar{D}_i^w$. The interaction channel carries over with the weighted switching leverage $\kappa_t^w = \sum_i w_{it} D_{i,t-1}(1 - \bar{D}_i^w)$ replacing κ_t. The pure-factor channel, however, no longer vanishes: the weighted cross-sectional sum $\xi_t = \sum_i w_{it} \tilde{D}_{i,t-1}^w$ need not be zero when weights vary over time. Since omitting factor f_k and its interaction removes exactly two regressors from the FE regression (f_{kt} and $D_{i,t-1} f_{kt}$), the value-weighted FE shift has the structure

$$\text{plim } \hat{\alpha}_{\text{FE}}^{\mathcal{A}'} - \text{plim } \hat{\alpha}_{\text{FE}}^{\mathcal{A}} = \underbrace{\Delta\beta_k \cdot \mu_{F_k}}_{\text{OLS shift}} + \underbrace{\Phi(S_\kappa^w, S_\xi)}_{\text{departure}}, \tag{11}$$

where

$$S_\kappa^w \equiv \sum_{t=1}^{T} \kappa_t^w \left(f_{kt} - \mu_{F_k}\right), \qquad S_\xi \equiv \sum_{t=1}^{T} \xi_t f_{kt},$$

and Φ is the multivariate omitted-variable departure that depends on S_κ^w and S_ξ through the joint covariance structure of the two omitted regressors with $\tilde{D}_{i,t-1}^w$ in the within-transformed regression. By standard OVB algebra, $\Phi(0,0) = 0$: when both S_κ^w and S_ξ vanish, the two omitted regressors are orthogonal to the within-transformed indicator, so the departure is zero and the value-weighted FE shift equals the OLS shift. In the equal-weight special case, $S_\xi = 0$ automatically and (11) reduces to (10). Condition F requires both $S_\kappa^w = 0$ and $S_\xi = 0$: restriction (6) eliminates the interaction departure; restriction (7) eliminates the pure-factor departure. Section 3.4 documents that the first restriction is directly well-supported in the data, and the empirical near-invariance of $\hat{B}^{\mathcal{A}}$ across factor models is consistent with the second restriction also being small.

Invariance. Under Condition F, the FE shift equals $\Delta\beta_k \cdot \mu_{F_k}$ for both equal and value weights, matching the OLS shift derived above. Subtracting one shift from the other gives $B^{\mathcal{A}'} - B^{\mathcal{A}} = \Delta\beta_k \cdot \mu_{F_k} - \Delta\beta_k \cdot \mu_{F_k} = 0$, establishing (8). □

Proof of Corollary 1 (Factor-Premium Adjustment of the Within-Firm Alpha).

The zero-factor FE regression has a single included regressor, the within-transformed portfolio indicator $\tilde{D}_{i,t-1} \equiv D_{i,t-1} - \bar{D}_i^w$ (using the weight-adjusted firm mean $\bar{D}_i^w$ under value weighting):

$$\tilde{r}_{it} \;=\; \alpha\, \tilde{D}_{i,t-1} \;+\; \tilde{\varepsilon}_{it}.$$

Moving from the zero-factor to the full specification $\mathcal{A}$ with K factors adds $2K$ within-transformed regressors: the factors $\tilde{f}_{kt}$ and the interactions $\tilde{Z}_{kit} \;\equiv\; D_{i,t-1}\, f_{kt} - \overline{D_i f_k}^{\,w}$, where the overbar denotes the weight-adjusted firm-specific time-series mean.

Because the zero-factor regression has a *single* included regressor, the multivariate omitted-variable formula gives

$$\operatorname{plim} \hat{\alpha}_{\mathrm{FE}}^{\varnothing} - \operatorname{plim} \hat{\alpha}_{\mathrm{FE}}^{\mathcal{A}} \;=\; \sum_{k=1}^{K} b_k \cdot \delta_{f_k} \;+\; \sum_{k=1}^{K} c_k \cdot \delta_{Z_k}, \tag{12}$$

where b_k and c_k are the population coefficients on $\tilde{f}_{kt}$ and $\tilde{Z}_{kit}$ in the full-factor FE regression, and

$$\delta_{f_k} \;\equiv\; \frac{\sum_{i,t} w_{it}\, \tilde{D}_{i,t-1}\, \tilde{f}_{kt}}{\sum_{i,t} w_{it}\, \tilde{D}_{i,t-1}^2}, \qquad \delta_{Z_k} \;\equiv\; \frac{\sum_{i,t} w_{it}\, \tilde{D}_{i,t-1}\, \tilde{Z}_{kit}}{\sum_{i,t} w_{it}\, \tilde{D}_{i,t-1}^2}.$$

The key property is that with a single included regressor in the short model, the multivariate OVB formula is a linear sum over the omitted regressors: cross-correlations among the $2K$ omitted terms affect the long-regression coefficients b_k and c_k but do not introduce additional cross-terms in (12). Orthogonalization of the factor basis is therefore not required.

Pure-factor channel. Due to $\tilde{f}_{kt}$ being common across firms within each period, $\sum_i w_{it}\, \tilde{D}_{i,t-1}\, \tilde{f}_{kt} \;=\; \tilde{f}_{kt}\, \xi_t$, where $\xi_t \;=\; \sum_i w_{it}\, \tilde{D}_{i,t-1}^w$ is the aggregate within-firm leverage from Condition F. Under equal weighting $\xi_t \;=\; 0$ for all t; under value weighting, Condition F restriction (7) sets $\sum_t \xi_t\, f_{kt} = 0$. In both cases $\delta_{f_k} = 0$.

Interaction channel. Using $D_{i,t-1} \in \{0, 1\}$ and the identity $\tilde{D}_{i,t-1}\, D_{i,t-1} \;=\; D_{i,t-1}(1 -$

$\bar{D}_i^w$):

$$\sum_{i,t} w_{it}\, \tilde{D}_{i,t-1}\, \tilde{Z}_{kit} \;=\; \sum_{i,t} w_{it}\, \tilde{D}_{i,t-1}\, D_{i,t-1}\, f_{kt} \;=\; \sum_{t=1}^{T} \kappa_t^w\, f_{kt},$$

where $\kappa_t^w = \sum_i w_{it}\, D_{i,t-1}(1 - \bar{D}_i^w)$ is the weighted switching leverage from Condition F, and the second equality uses $\sum_t w_{it}\, \tilde{D}_{i,t-1} = 0$ for each firm to absorb the firm-mean term. Since $\sum_{i,t} w_{it}\, \tilde{D}_{i,t-1}^2 = \sum_t \kappa_t^w$ (by the same binary-indicator identity used in the proof of Proposition 3):

$$\delta_{Z_k} \;=\; \frac{\sum_t \kappa_t^w\, f_{kt}}{\sum_t \kappa_t^w} \;=\; \omega_{F_k},$$

the switching-weighted factor mean. Under Condition F (6), $\omega_{F_k} = \mu_{F_k}$.

Long-regression coefficients. Under the standard exogeneity condition with respect to ε_{it}, the full-factor FE regression is consistently estimated, so the within-firm interaction coefficient satisfies $c_k \to \Delta\beta_k$, the structural differential factor loading.

Substituting into (12):

$$\operatorname{plim} \hat{\alpha}_{\mathrm{FE}}^{\varnothing} - \operatorname{plim} \hat{\alpha}_{\mathrm{FE}}^{\mathcal{A}} \;=\; \sum_{k \in \mathcal{A}} \Delta\beta_k \cdot \mu_{F_k},$$

which rearranges to (9). □

B Structural Decomposition of the Between-Firm Component

The results in Section 2 characterize B_0 without distributional assumptions. To obtain closed-form expressions that further separate the roles of sorting fineness, characteristic persistence, and the structural loading, this appendix introduces distributional structure on the sorting characteristic. Throughout this appendix, G denotes the population permanent-return gradient corresponding to B_0, that is, $G \equiv \mathrm{Cov}(\varphi_i, \bar{D}_i)/\mathrm{Var}(\bar{D}_i)$, the factor-free population limit of the specification-specific $G^{\mathcal{A}}$ introduced in Section 3.3.

Characteristic decomposition. Let $\mathcal{I}_i$ denote the sigma-field generated by firm i's permanent type. Define the permanent component of the sorting characteristic as the population conditional mean $\bar{z}_i \equiv E[z_{it} \mid \mathcal{I}_i]$, and the transitory component as the residual $\tilde{z}_{it} \equiv z_{it} - \bar{z}_i$. Then

$$z_{it} \;=\; \bar{z}_i \;+\; \tilde{z}_{it}, \tag{13}$$

where $\mathrm{Var}(\bar{z}_i) = \omega\,\sigma_z^2$, $\mathrm{Var}(\tilde{z}_{it}) = (1-\omega)\,\sigma_z^2$, and $\omega \in [0,1]$ is the between-firm variance share. By construction, $E[\tilde{z}_{it} \mid \mathcal{I}_i] = 0$.

Firm fixed effects. Since μ_i is $\mathcal{I}_i$-measurable and $E[\tilde{z}_{it} \mid \mathcal{I}_i] = 0$, the law of iterated expectations gives $\mathrm{Cov}(\mu_i, \tilde{z}_{it}) = E[\mu_i\,\tilde{z}_{it}] = E[\mu_i\,E[\tilde{z}_{it} \mid \mathcal{I}_i]] = 0$. We therefore project μ_i onto $\bar{z}_i$:

$$\mu_i \;=\; \delta \cdot \frac{\sigma_\mu}{\sigma_{\bar{z}}} \cdot \bar{z}_i \;+\; \eta_i, \qquad \eta_i \perp \bar{z}_i, \tag{14}$$

where $\delta = \mathrm{Corr}(\mu_i, \bar{z}_i)$ and $\sigma_\mu = \mathrm{sd}(\mu_i)$. Since $\eta_i = \mu_i - \delta(\sigma_\mu/\sigma_{\bar{z}})\bar{z}_i$ is a linear combination of $\mathcal{I}_i$-measurable objects, η_i is itself $\mathcal{I}_i$-measurable, and therefore $\mathrm{Cov}(\eta_i, \tilde{z}_{i,t-1}) = 0$ by the same law-of-iterated-expectations argument. The product $L \equiv \delta \cdot \sigma_\mu$, the **loading**, measures how strongly permanent firm heterogeneity projects onto the sorting characteristic.[16]

Corollary 3 (Structural Decomposition Under Normality). *Suppose $(\bar{z}_i, \tilde{z}_{i,t-1}, \mu_i)$ are **jointly normal**.[17] Consider symmetric sorts where H and L each contain the extreme $(1-q)$ fraction of firms. The permanence wedge from Section 3.3 decomposes multiplicatively as*

$$B_0 \;=\; \underbrace{2m_q}_{\text{sorting}} \;\cdot\; \underbrace{\sqrt{\omega}}_{\text{persistence}} \;\cdot\; \underbrace{L}_{\text{loading}}, \tag{15}$$

where $m_q = \phi(\Phi^{-1}(q))/(1-q)$ is the inverse Mills ratio measuring how far the average firm in the top group lies from the cross-sectional mean in standard-deviation units,[18] ω is the between-firm variance share (the fraction of characteristic variation attributable to permanent firm differences), $L = \delta \cdot \sigma_\mu$ is the loading measuring how strongly permanent firm heterogeneity projects onto the sorting characteristic, and $\delta = \mathrm{Corr}(\mu_i, \bar{z}_i)$ is the correlation between the permanent firm effect and the permanent component of the sorting characteristic. Asymmetry in the sort cutoffs or conditioning on additional characteristics would modify the closed-form constants but not the

[16] Under the normality assumption of Corollary 3, the loading relates to the permanent-return gradient introduced in Section 3.3 as follows: $G = L/\sigma_{\bar{z}}$, so the loading is the permanent-return gradient rescaled by the standard deviation of the permanent characteristic component.

[17] The result extends to the elliptical family, including the Student-t and logistic distributions, provided the distribution has finite second moments. The key property is that conditional expectations are linear, which is guaranteed for any elliptical distribution (see, e.g., Chamberlain, 1983; Fang and Zhang, 1990). Under heavier-tailed distributions, m_q exceeds the normal benchmark, implying stronger amplification of the between-firm component. Alternatively, the corollary holds under the weaker assumption that all conditional expectations among these variables are linear.

[18] For quintile sorts ($q = 0.8$), $m_{0.8} = 1.400$; for decile sorts ($q = 0.9$), $m_{0.9} = 1.755$. Note that m_q (the inverse Mills ratio, a function of the sorting cutoff, satisfying $m_q > 1$ for the quintile and decile cutoffs used in this paper) is distinct from λ (the within-firm variation share introduced in Section 3.3, always satisfying $\lambda \in [0,1]$).

logic of the decomposition.

Proof of Corollary 3. Under joint normality, conditional expectations are linear: $E[\bar{z}_i \mid z_{i,t-1}] = \omega \cdot z_{i,t-1}$. By symmetry, $E[z_{i,t-1}/\sigma_z \mid i \in H] = m_q$ and $E[z_{i,t-1}/\sigma_z \mid i \in L] = -m_q$, so the observed characteristic spread is $\Delta z = 2m_q\,\sigma_z$. Signal extraction gives $\Delta\bar{z} = \omega \cdot \Delta z$, and the standardized permanent spread is $\Delta\bar{z}/\sigma_{\bar{z}} = 2\sqrt{\omega}\,m_q$, where $\sigma_{\bar{z}} = \sqrt{\omega}\,\sigma_z$. From the projection (14), $E[\mu_i \mid H] - E[\mu_i \mid L] = (L/\sigma_{\bar{z}}) \cdot \Delta\bar{z} = 2m_q\sqrt{\omega}\,L$, provided $E[\eta_i \mid z_{i,t-1}] = 0$. This condition holds under the stated joint normality assumption. By the linear projection (14), $\mathrm{Cov}(\eta_i, \bar{z}_i) = 0$. Since η_i is $\mathcal{I}_i$-measurable, $\mathrm{Cov}(\eta_i, \tilde{z}_{i,t-1}) = 0$ by the law of iterated expectations (as shown above). Therefore η_i is uncorrelated with $z_{i,t-1} = \bar{z}_i + \tilde{z}_{i,t-1}$, and under joint normality uncorrelatedness implies $E[\eta_i \mid z_{i,t-1}] = 0$. $\square$

Correspondence with the product structure of Section 3.3. The assumption-free product structure $B_0 = (1-\lambda) \cdot G$ and the structural decomposition $B_0 = 2m_q\sqrt{\omega}\,L$ measure the same two economic forces in different coordinates: sorting permanence $(1-\lambda)$ and the product $2m_q\sqrt{\omega}$ both capture how persistently the sort tends to assign firms to the same portfolio; the permanent-return gradient G and the loading L both capture how strongly average expected returns vary with average portfolio assignment. The structural decomposition refines the assumption-free one by further separating sorting permanence into sorting fineness (m_q) and characteristic persistence ($\sqrt{\omega}$).

Sorting-granularity prediction and empirical test. The corollary yields a sharp testable prediction for the factor-free permanence wedge B_0. Because $B_0 = 2\,m_q\sqrt{\omega}\,L$, the ratio of B_0 under decile versus quintile sorts reduces to $m_{0.9}/m_{0.8} = 1.254$: decile sorts should amplify the permanence wedge by approximately 25%, a population-level result that depends only on the sorting-variable distribution. Since the estimated $\hat{B}^{\mathcal{A}}$ is near-invariant across the factor models considered (Section 3.4), the factor-free benchmark is informative for the factor-adjusted estimates as well. In sample, the cancellation is only approximate due to estimation noise and distributional departures; the corollary thus reduces the effect of sorting granularity on the permanence wedge to a single distributional parameter.

Ratio test. Table 9, Panel A reports the observed ratio $\hat{B}_{10}/\hat{B}_5$ across all anomalies for which the decile and quintile between-firm estimates share the same sign and the quintile denominator is non-degenerate (N ranges from 143 to 150 depending on the model, out of 171 total). Median ratios range from 1.495 (Market, $N = 144$) to 1.580 (FF 5F, $N = 150$). All four specifications reject H_0: $\hat{B}_{10}/\hat{B}_5 = 1$ with $p < 0.001$, documenting that sort granu-

Table 9: Sorting Granularity: Corollary 3 Tests and Significance Verdicts

Panel A: Ratio test $\hat{B}_{10}/\hat{B}_5$

Model	N	Median	IQR	p (Wilcoxon) H_0=1	p (Wilcoxon) H_0=1.254
Market	144	1.495	0.852	<0.001	<0.001
FF 3F	145	1.566	0.857	<0.001	<0.001
Carhart 4F	143	1.564	0.871	<0.001	<0.001
FF 5F	150	1.580	0.940	<0.001	<0.001

Panel B: Significance verdict (Market model)

Verdict	N	$\hat{B}$ sign >0	$\hat{B}$ sign <0	FE sig	$\hat{B}_{10}/\hat{B}_5$
Both significant	77	47	30	52	1.66
Decile only	17	6	11	11	1.52
Quintile only	8	2	6	5	0.90
Neither	69	30	39	24	1.37
Benchmark	171				1.254

Panel A: Two-sided Wilcoxon signed-rank tests of the between-firm ratio $\hat{B}_{10}/\hat{B}_5$ (decile vs. quintile VW sorts) across all 171 anomalies with valid same-sign, non-degenerate ratios ($|\hat{B}_5| \geq 10^{-8}$; N varies by model). H_0=1: sort granularity leaves $\hat{B}$ unchanged. H_0=1.254: Gaussian inverse Mills ratio benchmark $m_{0.9}/m_{0.8}$ predicted by Corollary 3. All four models reject both hypotheses ($p < 0.001$); observed medians 1.495–1.580 exceed the Gaussian benchmark by 19–26%, consistent with heavier-tailed sorting distributions. IQR: interquartile range of the ratio distribution. Panel B: OLS significance verdict for the Market model when switching between decile and quintile VW sorts ($|t_{\text{OLS}}| \geq 1.96$). $\hat{B}$ sign: >0 (inflation) or <0 (suppression). FE sig: $|t_{\text{FE}}| \geq 1.96$. $\hat{B}_{10}/\hat{B}_5$: within-cell median. Of the 77 anomalies classified as *Both significant*, 25 (32%) lack significant within-firm alpha; their OLS significance is driven by between-firm composition regardless of sort granularity. Specification: q10vw (decile sorts, value-weighted).

larity systematically amplifies the between-firm component. All four also reject the Gaussian benchmark H_0: $\hat{B}_{10}/\hat{B}_5 = 1.254$ ($p < 0.01$), with observed medians exceeding the benchmark by 19–26%. This systematic upward departure is directionally consistent with non-Gaussian sorting distributions: under fat-tailed distributions, the conditional tail expectation at the 90th percentile grows faster relative to the 80th than under the Gaussian, pushing the ratio above 1.254.[19] The broadly similar medians across Market (1.495), FF 3F (1.566), Carhart (1.564), and FF 5F (1.580) are consistent with the amplification factor depending on the sorting-variable distribution rather than on the factor specification, and with the near-invariance of $\hat{B}^{\mathcal{A}}$ across models documented in Section 3.4.

C Extension to Fama–MacBeth Regressions

The pooled-versus-within tension documented in the main text for portfolio sorts extends to the field's other canonical testing procedure. This appendix states and proves the formal result.

Proposition 4 (Equivalence of GPS Model and Fama–MacBeth Regressions)**.** *Weighted pooled OLS estimation of the GPS panel model with time fixed effects and period-specific control coefficients, using observation weights that assign equal total weight to each period, yields an estimate numerically identical to the Fama–MacBeth average slope, with identical HAC standard errors.*

Proof.

The standard Fama–MacBeth procedure estimates, for each period t, the cross-sectional regression

$$r_{it} \;=\; \gamma_{0t} \;+\; \gamma_{1t}\, z^*_{i,t-1} \;+\; \mathbf{c}'_{i,t-1}\, \boldsymbol{\phi}_t \;+\; u_{it}, \tag{16}$$

where $z^*_{i,t-1}$ is the cross-sectionally standardized characteristic (mean zero, unit standard deviation each period) and $\mathbf{c}_{i,t-1}$ is a vector of control characteristics. The FM estimator averages the slope coefficients: $\hat{\gamma}_{\text{FM}} = T^{-1} \sum_t \hat{\gamma}_{1t}$. The GPS panel model with time fixed effects is

$$r_{it} \;=\; \tau_t \;+\; \alpha \cdot z^*_{i,t-1} \;+\; \mathbf{c}'_{i,t-1}\, \boldsymbol{\phi}_t \;+\; \varepsilon_{it}, \tag{17}$$

estimated by weighted pooled OLS with observation weights $\omega_{it} = 1/W^c_t$, where $W^c_t = \sum_{i=1}^{N_t} (\tilde{z}^c_{i,t-1})^2$ is the cross-sectional sum of squared residual variation in the characteristic

[19]Under a Student-t distribution with five degrees of freedom, the analogous conditional tail expectation ratio is approximately 1.33; under $t(3)$, approximately 1.39.

after partialing out controls.

Part A (Coefficients). By the Frisch–Waugh–Lovell (FWL) theorem, the weighted pooled OLS slope on $z^*_{i,t-1}$ with observation weights $\omega_{it} = 1/W^c_t$ equals the slope from regressing the residual of r_{it} (after projection on τ_t and all control $\times$ time interactions) on the corresponding residual of $z^*_{i,t-1}$.

Since the control coefficients are period-specific, the projection operates within each period t independently: the residual of $z^*_{i,t-1}$ after projecting on $(1, \mathbf{c}_{i,t-1})$ within period t is exactly the FWL residual $\tilde{z}^c_{i,t-1}$ from the multivariate FM cross-section (16). Similarly, let $\tilde{r}^c_{it}$ denote the residual of r_{it} after the same within-period projection. With weights constant within each period, the weighted pooled OLS slope on the FWL residuals is:

$$\hat{\alpha}_{\text{WLS}} = \frac{\sum_{t=1}^{T}(1/W^c_t)\sum_{i=1}^{N_t}\tilde{z}^c_{i,t-1}\,\tilde{r}^c_{it}}{\sum_{t=1}^{T}(1/W^c_t)\,W^c_t} = \frac{\sum_{t=1}^{T}(1/W^c_t)\,W^c_t\,\hat{\gamma}_{1t}}{T},$$

where the last step uses the fact that the FM slope from the multivariate cross-section, by FWL, satisfies $\hat{\gamma}_{1t} = \sum_i \tilde{z}^c_{i,t-1}\,\tilde{r}^c_{it}/W^c_t$, so $\sum_i \tilde{z}^c_{i,t-1}\,\tilde{r}^c_{it} = W^c_t\,\hat{\gamma}_{1t}$. Therefore:

$$\hat{\alpha}_{\text{WLS}} = \frac{\sum_{t=1}^{T}\hat{\gamma}_{1t}}{T} = \hat{\gamma}_{\text{FM}},$$

establishing $\hat{\alpha}_{\text{WLS}} = \hat{\gamma}_{\text{FM}}$.

Part B (Standard Errors). The Driscoll–Kraay variance estimator for $\hat{\alpha}_{\text{WLS}}$ in the standard sandwich terminology is $\hat{V}_{\text{DK}} = \hat{\mathcal{B}}^{-1}\,\hat{\Omega}_H\,\hat{\mathcal{B}}^{-1}$, where $\hat{\mathcal{B}} = \sum_t (1/W^c_t)\,W^c_t = T$ is the "bread" and $\hat{\Omega}_H = \sum_{j=-H}^{H}\kappa(j/H)\sum_t g_t\,g_{t-j}$ is the Newey–West "meat" with kernel weights $\kappa(\cdot)$ and score contributions $g_t = \sum_i (1/W^c_t)\,\tilde{z}^c_{i,t-1}\,\hat{u}_{it}$, where $\hat{u}_{it}$ is the GPS residual evaluated at the FWL residuals.

Since $\hat{u}_{it} = \tilde{r}^c_{it} - \hat{\alpha}_{\text{WLS}}\,\tilde{z}^c_{i,t-1}$:

$$\begin{aligned} g_t &= \frac{1}{W^c_t}\sum_{i=1}^{N_t}\tilde{z}^c_{i,t-1}\big(\tilde{r}^c_{it} - \hat{\alpha}_{\text{WLS}}\,\tilde{z}^c_{i,t-1}\big) \\ &= \frac{1}{W^c_t}\Big(\sum_{i=1}^{N_t}\tilde{z}^c_{i,t-1}\,\tilde{r}^c_{it} - \hat{\alpha}_{\text{WLS}}\cdot W^c_t\Big) = \hat{\gamma}_{1t} - \hat{\gamma}_{\text{FM}}. \end{aligned}$$

Now consider the intercept-only regression of $\hat{\gamma}_{1t}$ on a constant. Its OLS estimator is $\hat{\gamma}_{\text{FM}} = T^{-1}\sum_t \hat{\gamma}_{1t}$ (the bread is T), and its residual is $\hat{e}_t = \hat{\gamma}_{1t} - \hat{\gamma}_{\text{FM}}$. The Newey–West score contribution is $g^{\text{OLS}}_t = \hat{e}_t = \hat{\gamma}_{1t} - \hat{\gamma}_{\text{FM}} = g_t$. Since the bread matrices (T

in both cases) and score sequences are identical, the sandwich variance estimators coincide: $\hat{V}_{\text{DK}} = \hat{V}_{\text{NW}}$, which is the standard Fama–MacBeth standard error with Newey–West HAC correction. $\square$

D Anomaly-Level GPS Classification

Table 10 lists the GPS decomposition for all 171 anomalies under the Market model. Panel A reports the 94 anomalies with significant sort alphas ($|t_{\text{OLS}}| \geq 1.96$), classified as Genuine (prediction alpha also significant, same sign) or Compositional (prediction alpha insignificant or opposite sign; significance driven by the permanence alpha). Panel B reports the remaining 77 anomalies, classified as Suppressed (sort alpha insignificant, prediction alpha significant; the permanence wedge masks a genuine within-firm effect) or Noise (neither significant).

Table 10: Anomaly-Level GPS Classification

Characteristic	Sort $\hat{\alpha}$ (bp/mo)	Prediction $\hat{\alpha}$ (bp/mo)	Permanence $\hat{B}$ (bp/mo)	$1-\lambda$
Panel A: OLS-Significant				
Genuine ($N = 62$)				
AbnormalAccruals	38.1***	38.9***	−0.8	0.503
Accruals	29.1**	32.0***	−2.9	0.606
AnalystRevision	67.3***	48.2***	19.2***	0.089
AnnouncementReturn	83.7***	73.6***	10.2***	0.107
AssetGrowth	49.3***	87.9***	−38.6***	0.402
BetaFP	−75.7***	−136.6***	60.8***	0.908
CPVolSpread	95.3***	78.6***	16.7	0.194
ChAssetTurnover	30.4***	38.5***	−8.1	0.171
ChEQ	53.5***	79.6***	−26.1***	0.401
ChInv	56.9***	68.9***	−12.0	0.317
ChInvIA	32.7***	38.8***	−6.1	0.340
ChTax	44.3***	28.7**	15.6***	0.116
CustomerMomentum	77.2***	59.8**	17.3	0.128
DelEqu	43.4***	80.1***	−36.7***	0.441
DelFINL	36.3***	35.5***	0.8	0.273
EarningsForecastDisparity	54.0**	44.6**	9.5	0.233
EntMult	42.9**	112.8***	−70.0***	0.787
EquityDuration	36.1**	40.1**	−4.0	0.765

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Table 10: Anomaly-Level GPS Classification (continued)

Characteristic	Sort $\hat{\alpha}$ (bp/mo)	Prediction $\hat{\alpha}$ (bp/mo)	Permanence $\hat{B}$ (bp/mo)	$1-\lambda$
IndRetBig	199.8***	189.0***	10.8*	0.155
InvGrowth	35.3***	38.2**	−2.9	0.278
InvestPPEInv	49.6***	61.6***	−12.1	0.446
Investment	48.7***	56.5***	−7.8	0.297
MaxRet	90.4***	73.0***	17.4***	0.620
Mom12m	137.9***	76.6***	61.3***	0.193
Mom12mOffSeason	122.5***	65.2***	57.3***	0.175
Mom6m	101.2***	56.0***	45.2***	0.104
MomOffSeason	38.6**	126.2***	−87.6***	0.453
MomOffSeason06YrPlus	38.2***	86.8***	−48.6***	0.490
MomOffSeason16YrPlus	27.3**	47.0***	−19.7*	0.433
MomSeason06YrPlus	76.5***	63.6***	12.9***	0.087
MomSeason11YrPlus	48.8***	41.5***	7.3	0.083
MomSeason16YrPlus	37.1***	38.4***	−1.3	0.085
NOA	63.3***	88.1***	−24.8	0.903
NetDebtFinance	26.2**	33.8***	−7.5**	0.346
NetPayoutYield	51.5***	38.5**	13.0	0.557
OPLeverage	35.0**	86.1**	−51.1**	0.996
OptionVolume2	27.3**	45.7***	−18.4***	0.152
REV6	60.4***	44.3**	16.1	0.180
RIVolSpread	76.2***	131.8***	−55.7***	0.250
ResidualMomentum	53.1***	43.9***	9.1***	0.073
ReturnSkew3F	−21.9***	−20.5***	−1.5**	0.052
ShareIss1Y	75.8***	60.0***	15.7**	0.505
ShareIss5Y	73.2***	60.4***	12.8	0.761
ShortInterest	54.4***	54.6***	−0.2	0.789
SmileSlope	138.4***	121.0***	17.3*	0.157
TrendFactor	129.7***	131.0***	−1.3	0.063
VolMkt	41.7**	54.3**	−12.7	0.916
VolSD	44.7***	151.7***	−107.0***	0.986
VolumeTrend	52.5***	73.8***	−21.3***	0.438
XFIN	77.6***	58.0***	19.7*	0.647
betaVIX	80.4***	71.8***	8.6***	0.078
cfp	54.6***	36.3**	18.3	0.561
dCPVolSpread	80.5***	85.9***	−5.4	0.031
dNoa	53.3***	71.2***	−17.9**	0.384
dVolCall	50.7**	62.4**	−11.7***	0.036
grcapx	45.0***	64.8***	−19.8*	0.332

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Table 10: Anomaly-Level GPS Classification (continued)

Characteristic	Sort $\hat{\alpha}$ (bp/mo)	Prediction $\hat{\alpha}$ (bp/mo)	Permanence $\hat{B}$ (bp/mo)	$1-\lambda$
grcapx3y	42.5***	67.5***	−24.9**	0.347
hire	31.9**	46.7***	−14.8	0.404
std_turn	125.4***	199.5***	−74.1**	0.919
zerotrade12M	53.6***	208.8***	−155.2***	0.908
zerotrade1M	44.2**	154.9***	−110.7***	0.838
zerotrade6M	52.4***	196.9***	−144.4***	0.890
Compositional ($N = 32$)				
Beta	−49.0**	25.4	−74.4***	0.954
BidAskSpread	−66.7***	−3.0	−63.7***	0.788
CBOperProf	74.6***	−26.3	100.9***	0.837
CompEquIss	44.1***	−17.2	61.2***	0.630
Coskewness	28.5**	26.5*	2.0	0.478
DelLTI	17.0**	15.6*	1.3	0.201
DelNetFin	21.7**	11.5	10.2	0.389
EarnSupBig	33.7**	27.5*	6.2	0.194
EarningsConsistency	33.2**	5.9	27.3	0.621
EarningsStreak	77.6***	31.1*	46.5***	0.363
EarningsSurprise	23.6**	16.8*	6.7***	0.064
FEPS	91.3***	52.9	38.3*	0.800
FirmAgeMom	180.7***	−85.8***	266.5***	0.417
ForecastDispersion	74.9***	52.6*	22.3	0.839
GP	39.4***	−17.5	56.9*	0.983
High52	43.3**	24.8	18.4***	0.163
IdioVol3F	92.6***	24.6	68.0***	0.771
IdioVolAHT	96.3***	46.0	50.3***	0.933
IntMom	74.3***	39.0*	35.2***	0.115
Mom6mJunk	125.7***	25.9	99.9***	0.226
MomSeason	38.5***	20.3	18.2***	0.081
MomSeasonShort	43.4**	27.7	15.7***	0.035
NetEquityFinance	50.8***	20.7	30.1***	0.716
OperProf	53.1***	26.3	26.8*	0.768
OperProfRD	69.8***	−36.6	106.5***	0.925
OrgCap	44.0***	20.1	23.9	0.961
PS	71.7**	105.1	−33.4	0.725
PctTotAcc	28.7**	13.8	14.9	0.399
RealizedVol	102.2***	54.0*	48.2***	0.755
Tax	40.1***	20.0	20.2	0.664

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Table 10: Anomaly-Level GPS Classification (continued)

Characteristic	Sort $\hat{\alpha}$ (bp/mo)	Prediction $\hat{\alpha}$ (bp/mo)	Permanence $\hat{B}$ (bp/mo)	$1-\lambda$
roaq	53.1***	−26.6	79.7***	0.672
sfe	83.7***	−30.6	114.4**	0.868
Panel B: OLS-Insignificant				
Suppressed ($N = 29$)				
AM	5.9	217.0***	−211.1***	0.981
AdExp	18.1	370.8***	−352.8*	0.983
BM	26.7	142.0***	−115.3***	0.943
BMdec	28.3	157.4***	−129.1***	0.923
CF	40.7*	79.5***	−38.8	0.762
CashProd	25.0	84.0***	−59.0**	0.876
ChNNCOA	19.5*	21.4**	−1.8**	0.193
DelCOA	16.4	42.9***	−26.6**	0.459
DolVol	20.9	349.2***	−328.3***	0.990
EP	20.3	44.1**	−23.8	0.707
Frontier	6.2	144.6***	−138.4***	0.807
GrAdExp	26.0	38.6**	−12.6	0.427
Illiquidity	12.6	348.3***	−335.7***	0.995
IntanBM	2.6	74.8***	−72.2***	0.688
IntanCFP	7.7	61.9**	−54.1***	0.561
LRreversal	12.4	69.9***	−57.4***	0.312
Leverage	3.0	274.4***	−271.5***	0.991
MRreversal	35.8*	73.5***	−37.7***	0.115
MeanRankRevGrowth	12.7	41.2**	−28.6*	0.626
MomOffSeason11YrPlus	13.6	36.3**	−22.7***	0.445
OptionVolume1	38.2	267.5***	−229.3***	0.942
PriceDelayRsq	10.8	53.4***	−42.6***	0.752
ProbInformedTrading	45.7	290.4**	−244.7*	0.828
RDS	25.0*	32.0***	−7.1	0.201
SP	40.4*	280.4***	−240.0***	0.955
Size	−3.3	−445.5***	442.2***	0.996
TotalAccruals	20.1*	41.9***	−21.8***	0.337
VarCF	3.2	−62.6**	65.8***	0.898
fgr5yrLag	42.3*	81.0**	−38.7**	0.798
Noise ($N = 48$)				
AOP	24.7	32.4*	−7.7	0.490
Activism1	−13.0	−81.4	68.4***	0.992

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Table 10: Anomaly-Level GPS Classification (continued)

Characteristic	Sort $\hat{\alpha}$ (bp/mo)	Prediction $\hat{\alpha}$ (bp/mo)	Permanence $\hat{B}$ (bp/mo)	$1-\lambda$
AnalystValue	18.4	−18.2	36.6	0.709
BPEBM	−1.0	16.2	−17.3***	0.270
BetaLiquidityPS	1.4	29.1*	−27.6	0.612
BetaTailRisk	12.7	57.1	−44.4	0.922
BookLeverage	−3.7	−5.7	1.9	0.920
BrandInvest	−16.5	−45.5	29.0	0.784
Cash	17.8	0.7	17.1	0.940
ChNWC	19.1*	19.8*	−0.6	0.198
ChangeInRecommendation	6.7	2.3	4.4*	0.034
CompositeDebtIssuance	−1.5	−4.5	3.0	0.470
CoskewACX	25.5*	9.0	16.5***	0.229
DelBreadth	22.9	10.5	12.3***	0.069
DelCOL	2.5	23.1*	−20.6	0.418
DelDRC	−20.4	−25.2	4.8	0.572
EBM	10.2	24.4*	−14.2***	0.304
ExclExp	12.7	−13.1	25.9	0.173
FR	16.2	−6.8	23.0	0.737
GrLTNOA	14.2	7.8	6.4	0.529
GrSaleToGrInv	18.3	17.7	0.7	0.242
GrSaleToGrOverhead	−9.2	−21.1	11.9	0.278
Herf	0.0	−0.1	0.2	0.949
HerfAsset	−4.6	−41.8	37.2	0.965
HerfBE	3.9	7.6	−3.7	0.971
IndMom	36.1*	23.1	13.0**	0.141
IntanEP	0.9	37.7*	−36.7***	0.457
IntanSP	−26.8	42.4	−69.1***	0.536
NetDebtPrice	5.4	18.2	−12.9	0.829
OrderBacklog	−20.9	−90.4*	69.4	0.962
OrderBacklogChg	−18.9	−4.8	−14.2	0.139
PayoutYield	33.4*	41.5*	−8.1	0.735
PctAcc	21.0	6.4	14.6	0.518
PredictedFE	0.0	−13.6	13.6	0.689
PriceDelaySlope	−1.7	4.9	−6.6***	0.469
PriceDelayTstat	−2.6	4.2	−6.8***	0.468
RD	86.9	254.0	−167.1*	0.994
RDAbility	20.8	−10.2	31.0	0.562
ReturnSkew	−8.7	−9.9	1.2*	0.056
RevenueSurprise	15.7	21.8*	−6.1	0.106

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Table 10: Anomaly-Level GPS Classification (continued)

Characteristic	Sort $\hat{\alpha}$ (bp/mo)	Prediction $\hat{\alpha}$ (bp/mo)	Permanence $\hat{B}$ (bp/mo)	$1-\lambda$
RoE	26.7*	−13.3	40.0***	0.660
dVolPut	14.8	3.1	11.7***	0.037
iomom_cust	27.4	26.9	0.5	0.099
iomom_supp	22.1	14.6	7.5	0.134
realestate	15.8	−36.0	51.8	0.931
retConglomerate	35.1*	32.7	2.4	0.127
skew1	39.9*	1.5	38.4**	0.457
tang	11.0	56.7	−45.7*	0.934

GPS decomposition of all 171 anomalies with valid GPS estimates under the Market model (value-weighted decile sorts). Sort $\hat{\alpha}$: conventional portfolio-sort alpha $\hat{\alpha}^{\mathcal{A}}_{\text{OLS}}$. Prediction $\hat{\alpha}$: within-firm alpha $\hat{\alpha}^{\mathcal{A}}_{\text{FE}}$. Permanence $\hat{B}$: between-firm component $\hat{B}^{\mathcal{A}} \equiv \hat{\alpha}^{\mathcal{A}}_{\text{OLS}} - \hat{\alpha}^{\mathcal{A}}_{\text{FE}}$. $1-\lambda$: sorting permanence. Classification: Genuine = both sort and prediction alpha significant at 5% (same sign); Compositional = sort alpha significant, prediction alpha not (or reverses sign); Suppressed = sort alpha insignificant, prediction alpha significant; Noise = neither significant. Panel A lists the 94 anomalies with significant sort alphas ($|t| \geq 1.96$): 62 retain significance under the prediction alpha (Genuine), while 32 (34%) do not (Compositional). Panel B lists the 77 anomalies with insignificant sort alphas: 29 have significant prediction alphas (Suppressed). Significance stars for the sort and prediction alphas are based on Driscoll–Kraay standard errors (lag = 3); significance for the permanence bias is based on the Hausman (1978) specification test of H_0: $\hat{B}^{\mathcal{A}} = 0$. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Figure 3 provides a visual complement to Table 10, displaying the anomaly-level GPS decomposition across all four factor models.

Figure 3: Cross-Model Stability of GPS Components

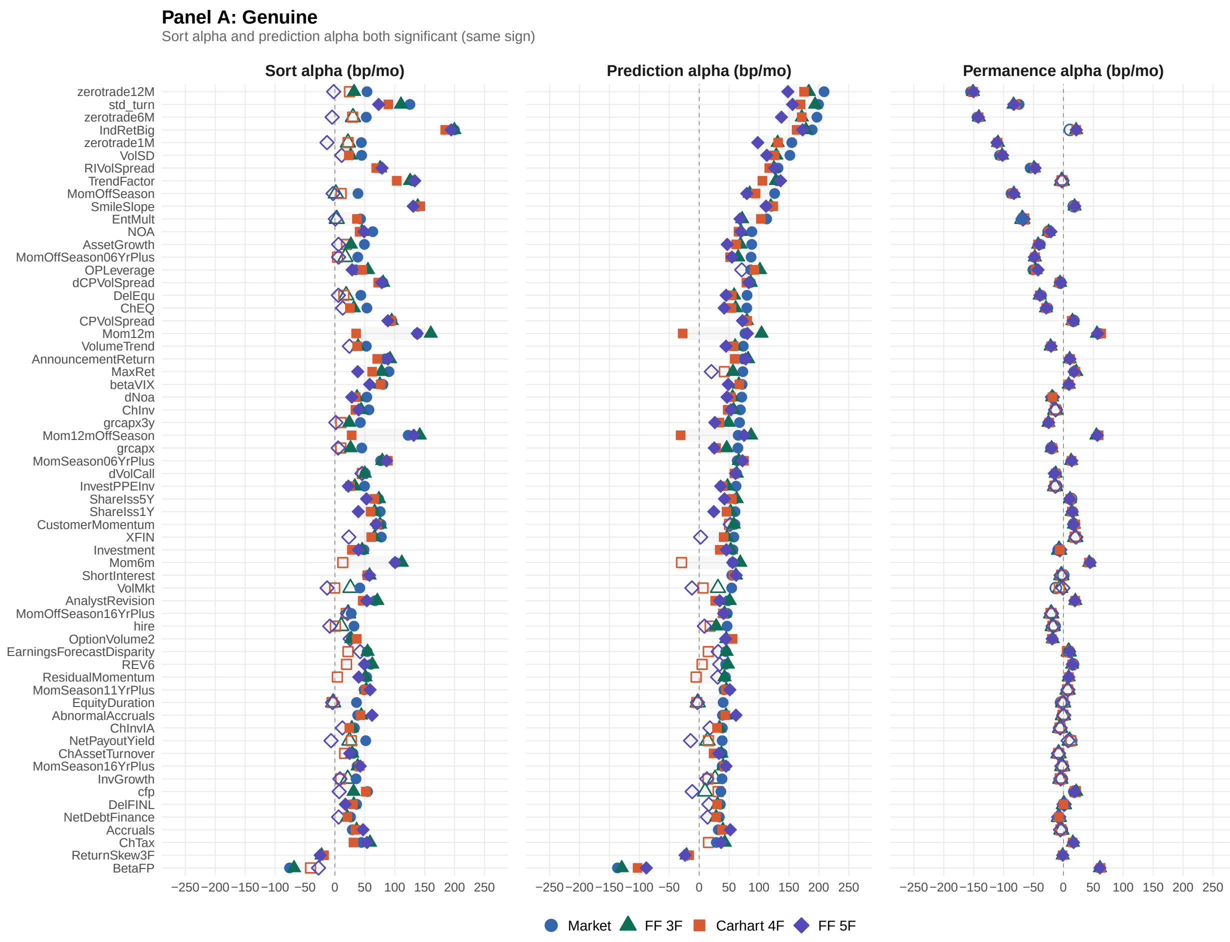


Figure 3: Cross-Model Stability of GPS Components (continued)

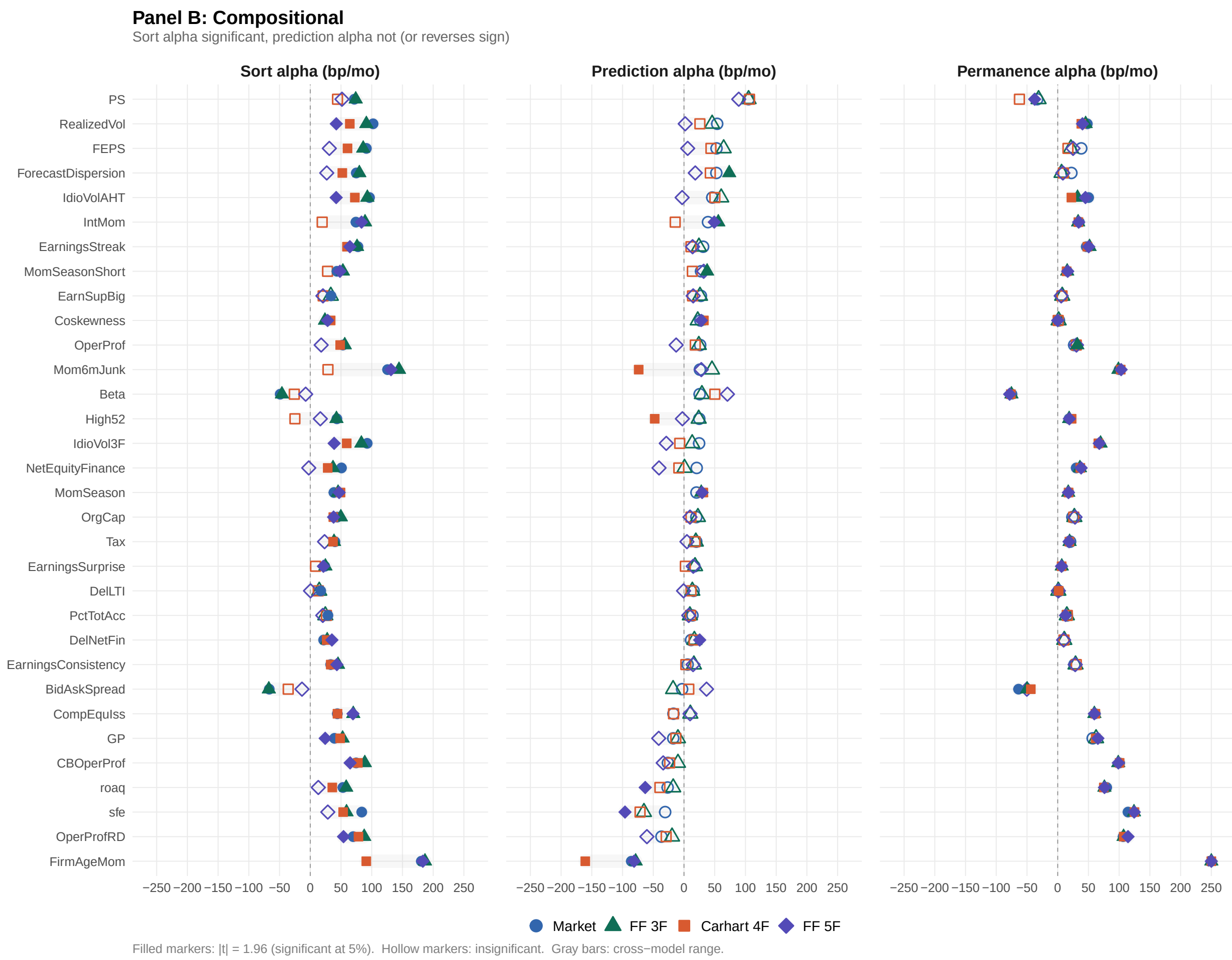


Figure 3: Cross-Model Stability of GPS Components (continued)

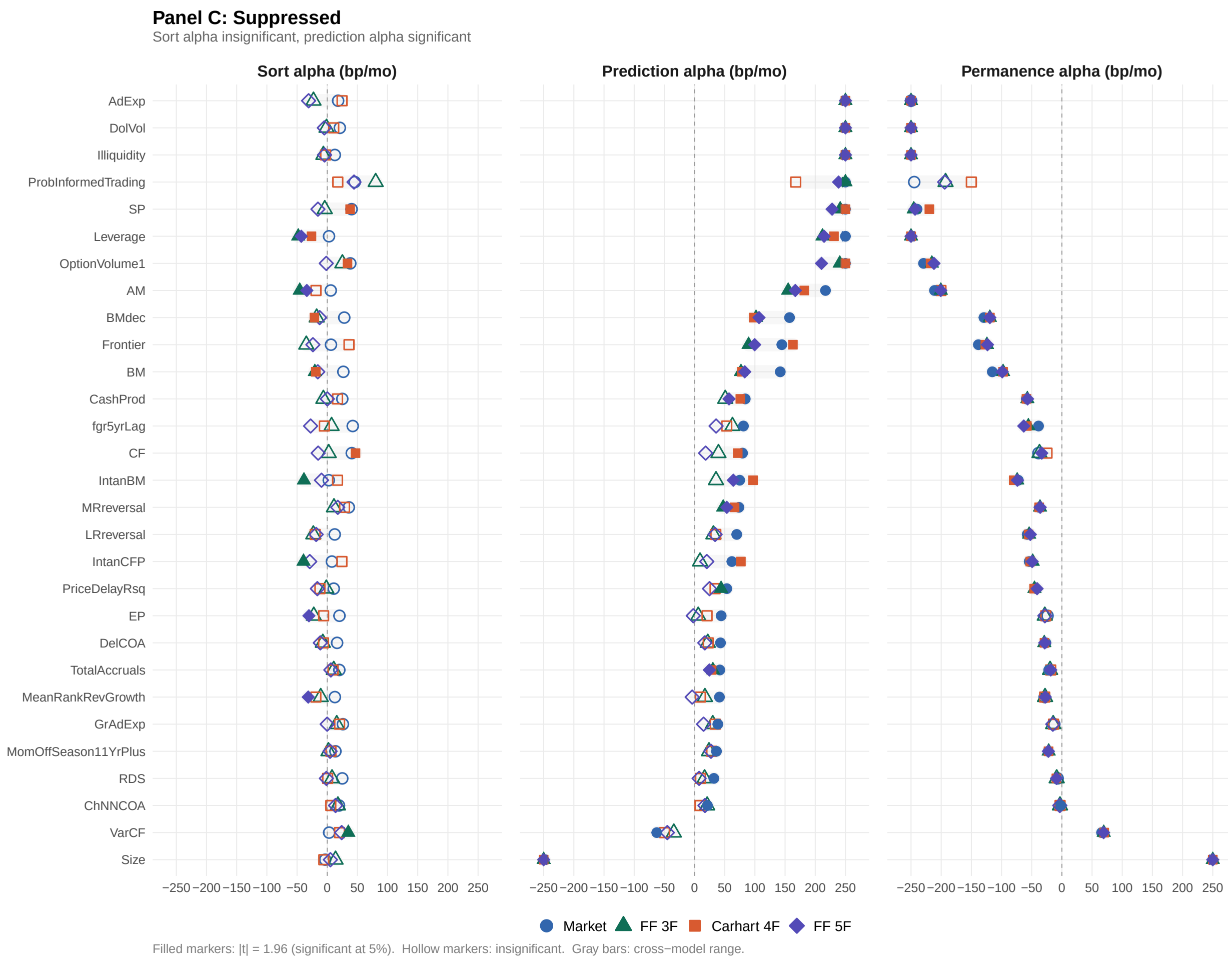
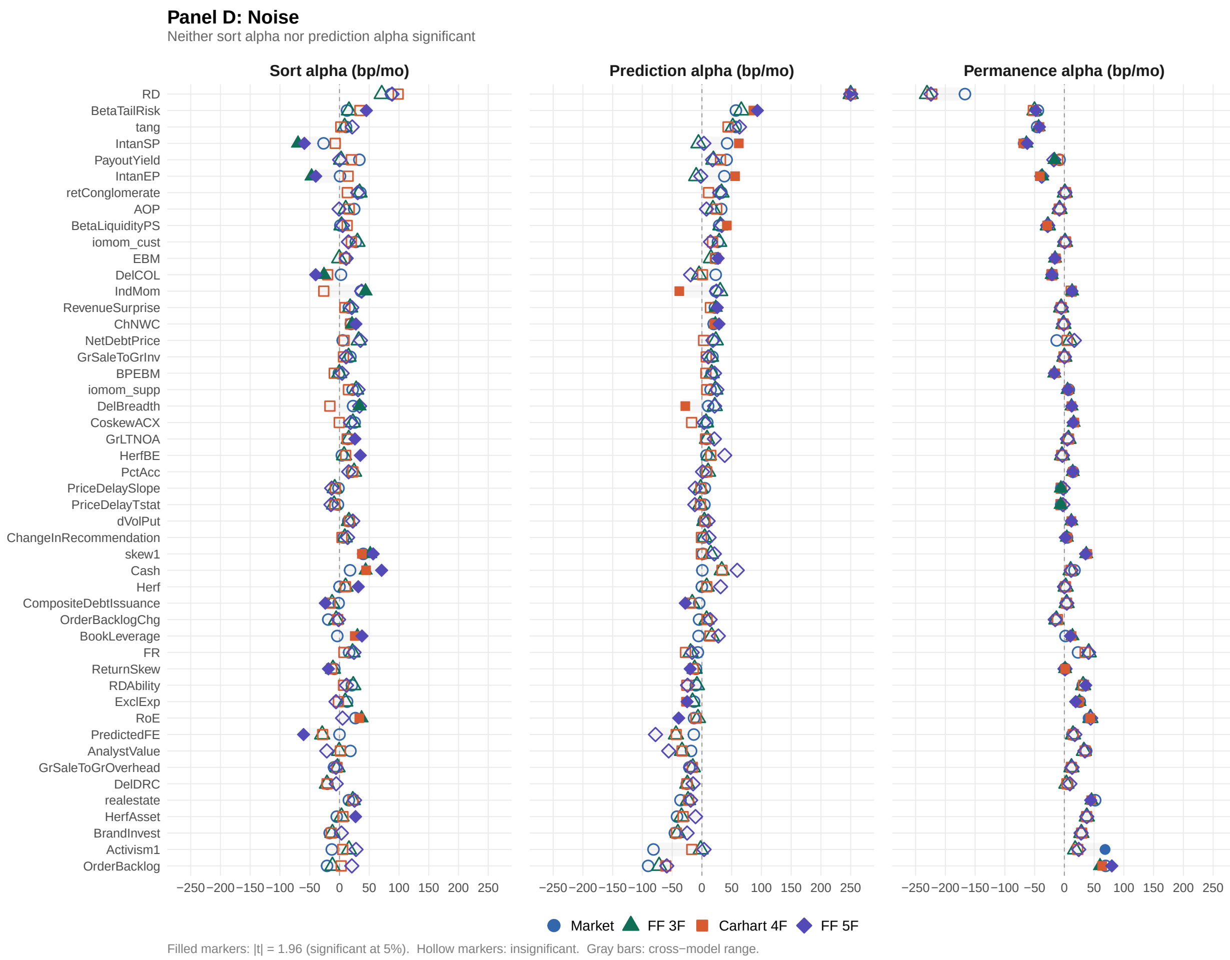


Figure 3: Cross-Model Stability of GPS Components (continued)



This figure plots the GPS decomposition for each of the 171 anomalies across four factor models: Market, FF 3F, Carhart, and FF 5F. Each row represents one anomaly; the three columns show the sort alpha ($\hat{\alpha}_{\text{OLS}}$), prediction alpha ($\hat{\alpha}_{\text{FE}}$), and permanence wedge ($\hat{B}^{\mathcal{A}}$), all in basis points per month. Marker shapes distinguish factor models (circle = Market, triangle = FF 3F, square = Carhart, diamond = FF 5F); filled markers denote statistical significance at the 5% level ($|t| \geq 1.96$). Gray bars span the cross-model range. Anomalies are grouped by GPS classification (Panel A: Genuine; Panel B: Compositional; Panel C: Suppressed; Panel D: Noise) and sorted within each panel by Market-model prediction alpha in descending signed order. The x-axis is truncated at ± 250 bp/month; off-chart values are indicated by arrows.

E Multi-Characteristic Robustness

Four-characteristic extension. Table 11 adds twelve-month momentum to the three-characteristic specification in Table 8. This extension serves two purposes: it sharpens the dimensionality argument by moving from 125 to 625 conventional sort cells, and it introduces a return-based characteristic whose behavior contrasts with the slow-moving accounting variables in the three-characteristic case. The main conclusions from the three-characteristic specification survive: book-to-market remains suppressed ($\hat{B} = -35.2$ bp), gross profitability remains conditioning-sensitive ($\hat{B} = +7.4$ bp), and investment continues to display only a minimal permanence wedge ($\hat{B} = -1.3$ bp). Momentum itself displays a distinct pattern: its sort alpha of 34.3 bp/month ($t = 1.99$) is almost entirely attributable to the permanence wedge (+29.5 bp), with a prediction alpha of only 4.8 bp/month ($t = 0.28$). This composition-driven pattern is consistent with the transient nature of past-return characteristics (Section 3.3). We interpret this result as illustrative rather than definitive, since momentum is a return-based characteristic for which the standard exogeneity condition is an identifying condition rather than a mechanical consequence of lagging.

Table 11: GPS with Four Characteristics

	Univariate continuous GPS			Multi-characteristic GPS		
	OLS $\hat{\alpha}$	Pred $\hat{\alpha}$	$\hat{B}$	OLS $\hat{\alpha}$	Pred $\hat{\alpha}$	$\hat{B}$
Characteristic	(bp)	(bp)	(bp)	(bp)	(bp)	(bp)
Book-to-market	0.8	42.6***	−41.8	8.0	43.2***	−35.2
Gross profitability	24.5*	3.7	20.8	26.0**	18.6	7.4
Investment	13.8	13.9	−0.1	10.4	11.6	−1.3
Momentum	36.3**	7.4	28.9	34.3**	4.8	29.5

Four-characteristic GPS decomposition under the Market model, adding twelve-month momentum to the three-characteristic specification in Table 8. All conventions as in Table 8. Sample: 1,544,255 firm-month observations (12,362 firms). Significance based on Driscoll–Kraay standard errors (lag = 3). *, **, ***: 10%, 5%, 1% levels.

Robustness across factor models. Table 12 repeats the multi-characteristic GPS analysis under the Carhart and Fama–French five-factor models. The main economic conclusions are unchanged. In the three-characteristic specification, book-to-market remains a suppressed anomaly across all benchmarks: its sort alpha is small, its prediction alpha remains large, and the permanence alpha remains strongly negative. Gross profitability continues to be conditioning-sensitive: relative to the corresponding univariate continuous specification, joint conditioning raises the prediction alpha and reduces the permanence wedge. Investment continues to exhibit only a minimal permanence wedge.

The same pattern carries over to the four-characteristic extension. Book-to-market remains suppressed, gross profitability remains conditioning-sensitive, and investment continues to display little separation between the sort alpha and the prediction alpha. Momentum's estimated alpha remains concentrated primarily in the permanence wedge, while its prediction alpha is small across specifications. We treat the Carhart and five-factor results as robustness checks rather than the primary benchmark in the joint setting, because the included factors overlap with the sorting characteristics. Their main role is to show that the multi-characteristic decomposition is not specific to the Market model and that the permanence wedge remains highly stable across specifications.

Table 12: Multi-Characteristic GPS: Robustness Across Factor Models

	Univariate continuous GPS			Multi-characteristic GPS		
	OLS $\hat{\alpha}$	Pred $\hat{\alpha}$	$\hat{B}$	OLS $\hat{\alpha}$	Pred $\hat{\alpha}$	$\hat{B}$
Characteristic	(bp)	(bp)	(bp)	(bp)	(bp)	(bp)
Panel A: BM, GP, Investment: Carhart model						
Book-to-market	−3.4	39.0***	−42.4	4.5	42.2***	−37.7
Gross profitability	33.8***	14.7	19.1	34.1***	27.9	6.3
Investment	6.8	6.6	0.2	6.0	4.8	1.3
Panel B: BM, GP, Investment: FF 5F model						
Book-to-market	−2.3	40.4***	−42.7	3.7	41.2***	−37.5
Gross profitability	28.1***	8.0	20.1	28.6***	21.3	7.3
Investment	7.9	7.9	0.1	6.6	5.7	0.9
Panel C: BM, GP, Investment, Momentum: Carhart model						
Book-to-market	−3.4	39.0***	−42.4	5.8	41.5***	−35.7
Gross profitability	33.8***	14.6	19.2	32.3***	25.5	6.9
Investment	6.9	6.6	0.3	7.5	8.3	−0.8
Momentum	8.6	−18.1	26.7	7.5	−19.7	27.3
Panel D: BM, GP, Investment, Momentum: FF 5F model						
Book-to-market	−2.3	40.5***	−42.7	4.0	39.2***	−35.2
Gross profitability	28.0***	7.9	20.1	24.2***	18.9	5.3
Investment	8.0	7.9	0.1	2.6	3.2	−0.6
Momentum	43.7***	12.0	31.6	41.5***	10.0	31.5

Robustness of the multi-characteristic GPS analysis across factor models. Table format matches Table 8; the Market model results from that table are omitted here. Within each panel, univariate and multi-characteristic columns use the same sample and benchmark; the only difference is joint conditioning. Significance based on Driscoll–Kraay standard errors (lag = 3). *, **, ***: 10%, 5%, 1% levels.